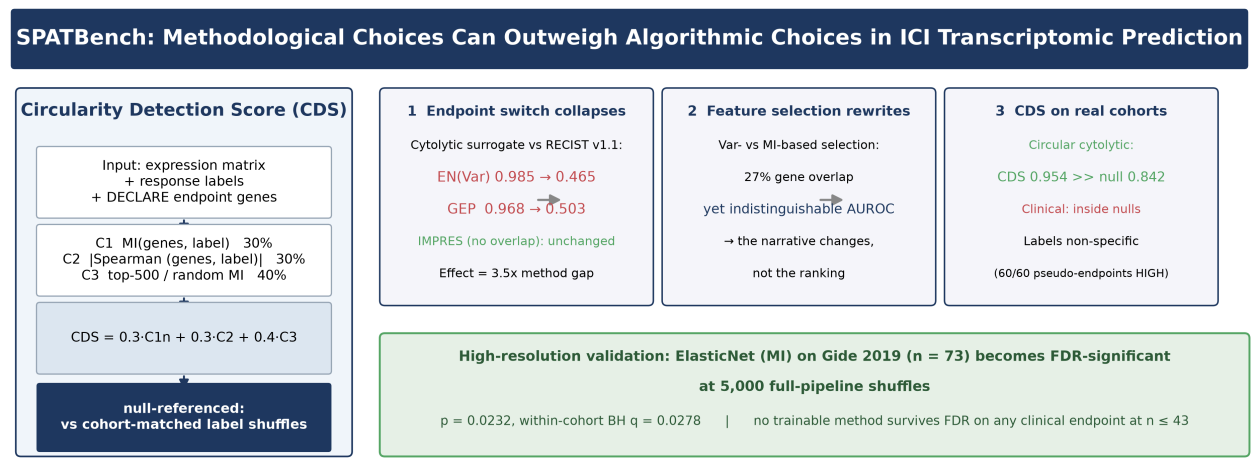


SPATBench: Methodological Choices Can Outweigh Algorithmic Choices in Transcriptomic Immunotherapy Prediction

Benchmarking at Current Sample Sizes



Graphical abstract. CDS workflow (left) and the three main findings (right); bottom: high-resolution validation on Gide 2019.

Zhixian Huang (黄志贤)¹

¹ Putian University, Putian, Fujian, China

Lead contact: Zhixian Huang, huangzhixian324@gmail.com (correspondence)

Author affiliations: ¹ Putian University, Putian, Fujian, China

SUMMARY

Transcriptomic predictors of immune checkpoint inhibitor (ICI) response are benchmarked on small, incompatible cohorts with inconsistent endpoints, obscuring whether conclusions reflect algorithms or design. SPATBench isolates three such design choices across six methods and five immunotherapy-treated cohorts (196 patients; 169 melanoma). Switching from a molecular surrogate (cytolytic activity) to a clinical endpoint (RECIST v1.1) in the same patients (melanoma, Riaz) collapses every method whose features overlap the endpoint-defining genes (ElasticNet-Var 0.985 to 0.465; GEP 0.968 to 0.503), a larger effect than the largest within-cohort method gap, while two clinical definitions on identical patients move predictions by at most 0.084, and the effect replicates in executed external cohorts spanning five cancer types. MI- and variance-based selection yield 27%-overlapping gene sets and indistinguishable performance. The Circularity Detection Score (CDS) flags the circular endpoint far above its null, yet its categorical labels prove non-specific. No trainable method survives FDR correction at $n \leq 43$, whereas at $n = 73$ (Gide) ElasticNet-MI is significant within its cohort family ($p = 0.0232$). Both tools are open-source and Docker-reproducible.

INTRODUCTION

Immune checkpoint inhibitors (ICIs) targeting PD-1, PD-L1, and CTLA-4 have transformed oncology [1,2], yet the transcriptomic biomarkers developed to guide their use rest on a methodological foundation that has not itself been systematically audited. Only 20–30% of unselected patients achieve objective responses, motivating intensive efforts to develop transcriptomic biomarkers for stratification [3–6]. Multiple gene-expression-based prediction methods have been proposed: the T-effector gene expression profile (GEP) [3], TIDE [4], IMPRES [5], and a growing number of machine learning (ML) models [6–8]. Each was evaluated on different cohorts, preprocessing, response definitions and protocols — a fragmentation that produced a literature in which nearly every method appears superior, but only within its own developmental context [9,10]. Three structural features of the field make method rankings unreliable: (i) public immunotherapy cohorts with pre-treatment RNA-seq are small (typically $n < 50$), limiting power [11–13]; (ii) response can be defined via clinical endpoints (RECIST v1.1), molecular surrogates (cytolytic activity [14]), or progression-based endpoints (PFS), each capturing different biology; and (iii) methodological choices such as feature selection are rarely treated as experimental variables. Several recent efforts have benchmarked immunotherapy prediction methods across multiple cohorts [15–18]. Liang et al. [15] found limited cross-cohort generalizability for nine predictors, COMPASS [16] cross-cancer improvements on 1,133 patients, the ICLR 2026 benchmark [17] domain-specific models outperforming general-purpose ones across 35 tasks, and EXPRESSO [18] 20 signatures across 3,729 patients; a re-analysis of 17 tumor-microenvironment scoring methods found none reliable across cancer types [19] (Supplementary Table S2 compares their published AUROCs with ours). All hold methodology fixed and compare methods; our approach is orthogonal — we hold methods fixed and systematically vary methodological choices, asking which design decisions, rather than which algorithms, determine benchmark outcomes. Here we present SPATBench, a computational framework and benchmark resource asking: *given that rankings are unreliable at available sample sizes, which methodological choices contribute most to that unreliability, and by how much?* Our contributions are:

1. An open-source benchmark framework enforcing patient-stratified cross-validation and standardized preprocessing across six prediction methods and five cohorts.
2. Empirical isolation and quantification of three design choices — response definition, feature selection, and endpoint circularity, in which the endpoint is defined by genes that also appear in the predictor's features so the evaluation loses independence — with effect-size comparisons against method-level variation.
3. A Circularity Detection Score (CDS) for pre-benchmarking endpoint assessment — a null-referenced continuous diagnostic whose categorical labels are advisory only; the validated deliverable is a reporting convention whose binding rule is the declaration of endpoint-defining genes, with CDS as its quantitative companion rather than an independent gate — evaluated on 5 real endpoints, 5 synthetic scenarios, 60 pseudo-endpoints, one retrospective external consistency check (Auslander et al. 2018), and published benchmark studies.
4. A complete permutation test matrix with high-resolution validation for borderline cases, showing that no trainable method is distinguishable from label shuffling on any clinical endpoint at $n \leq 43$ after FDR correction.
5. A power-analysis-derived checklist of minimum benchmarking standards. All evidence is retrospective on public cohorts; the response-definition result is demonstrated in five cancer types

(melanoma, bladder carcinoma, NSCLC, renal cell carcinoma and gastric cancer) via executed external replications.

RESULTS

Design Choice 1: Response Definition Exerts a Larger Effect Than Method Choice

The Riaz 2017 cohort provides two response definitions on the same patients and expression matrix.

Table 1. Riaz 2017: RECIST (n = 42) versus cytolytic (n = 43) response definitions (v33, HGNC-mapped).

Method	RECIST AUROC (n = 42)	Cytolytic AUROC (n = 43)	Δ AUROC	Type
ElasticNet-MI	0.569 [0.370–0.755]	0.968 [0.909–1.000]	−0.399	ML
ElasticNet-Var	0.465 [0.208–0.704]	0.985 [0.954–1.000]	−0.520	ML
TIDE	0.466 [0.222–0.723]	0.470 [0.282–0.653]	−0.004	Signature
IMPRES	0.613 [0.381–0.835]	0.660 [0.493–0.811]	−0.047	Signature
GEP	0.503 [0.305–0.710]	0.968 [0.915–1.000]	−0.465	Signature
PD-L1 (CD274)	0.545 [0.330–0.760]	0.868 [0.750–0.960]	−0.323	Biomarker

Switching response definitions transforms all ML methods from apparently state-of-the-art (Var 0.985 on cytolytic) to near-chance on RECIST (0.465; Figure 1); GEP collapses with them (0.968 $\rightarrow$ 0.503) because the cytolytic endpoint is defined by GZMA/PRF1, both members of the GEP signature. IMPRES — whose fifteen checkpoint-gene relations share no genes with the endpoint definition — is nearly insensitive (0.660 $\rightarrow$ 0.613, Δ = −0.047; permutation p = 0.032 on cytolytic, 0.154 on RECIST, each against its own null). Within the Riaz RECIST endpoint the largest between-method Δ AUROC is 0.147 (IMPRES vs TIDE); the largest between-endpoint effect (0.520) exceeds it 3.5 $\times$ (Supplementary Table S14). The 0.520 contrast uses the benchmark (seed-42) split; across 10 stratified seeds the k = 2 RECIST mean is 0.389 $\pm$ 0.105 (Supplementary Table S16) — the collapse carries $\approx$ 0.1 seed SD while its direction and scale are stable. **k-fold robustness:** The Riaz RECIST endpoint uses only 2-fold CV (9 responders). Re-running ElasticNet-Var with k = 3 and k = 5 (10 stratified seeds per k) keeps all RECIST AUROC below 0.60 (k = 3: 0.397 $\pm$ 0.064; k = 5: 0.398 $\pm$ 0.088; Supplementary Table S16; results/benchmark/v33/riaz_recist_kfold_v33.json) — not an artifact of the k = 2 constraint.

External replication (pre-declared, i.e. its protocol was fixed before the replication data were analyzed): The effect replicates on an independent cohort — Liu 2019 ^[20] (n = 122 annotated samples; DFCI anti-PD-1 melanoma; independent of all five benchmark cohorts; iAtlas harmonization; the first executed external replication). On the same patients and expression matrix, the cytolytic-surrogate endpoint (GZMA/PRF1 median split — the Riaz construction) yields GEP 0.959 and ElasticNet-MI 0.956, RECIST 0.535 and 0.538 (Δ = +0.425 and +0.418); IMPRES, whose features exclude the endpoint genes, retains the most RECIST signal (0.601) and is the only nominally significant fixed scorer (p = 0.027; Benjamini-Hochberg q = 0.108), mirroring the Hugo headline result; no trainable CI excludes 0.5 on RECIST. CDS flags the surrogate HIGH (0.949 vs null mean 0.566) and places the clinical endpoint above

its null mean (0.721 vs 0.591) — the Gide genuine-biology signature (Table S4b). Full results in Supplementary Table S19; pipeline in scripts/e1_replication_liu2019.py.

Pre-declared NSCLC candidate executed: GSE274975 [21] (anti-PD-(L)1 NSCLC, n = 61; RECIST responders 15) reports its RECIST and PFS endpoints in the source publication's supplementary table rather than in GEO metadata. After recovering them and verifying sample identity (patient-number bijection across the counts matrix, the GEO titles and the clinical table; histotype concordance 57/58), the pre-declared analysis protocol ran unchanged. RECIST leaves PD-L1 expression as the only fixed scorer surviving label permutation (0.690 [0.513–0.865], perm p = 0.014, BH q = 0.056), consistent with PD-L1 IHC being the clinically approved biomarker in this indication; GEP reaches 0.613 (p = 0.100) and IMPRES, TIDE and both ElasticNets sit at chance. The surrogate endpoint inflates GEP to 0.881 (p = 2×10^{-5} , BH q = 8×10^{-5} ; CDS 0.786 HIGH). Full results in Supplementary Table S27; pipeline in scripts/e1_gse274975_executed.py.

Cross-cancer replication (pre-declared): The effect replicates in a non-melanoma tumor type — IMvigor210 [22] (metastatic urothelial carcinoma, n = 297 response-annotated samples; anti-PD-L1; iAtlas harmonization). On identical patients, the cytolytic-surrogate endpoint yields GEP 0.967 and ElasticNet-MI 0.951, RECIST 0.613 and 0.657 ($\Delta = +0.354$ and $+0.294$); PD-L1 shows the same pattern (0.863 $\rightarrow$ 0.566, $\Delta = +0.297$). On the clinical endpoint, GEP is the only fixed scorer surviving FDR (p = 0.0025; BH q = 0.010) — a weak effect (AUROC 0.613) detectable only at this sample size (Power Analysis); IMvigor210 is an application cohort of the GEP [3], so this significance reflects cross-cohort signal, not cohort of origin. CDS separates the two endpoints in this cohort: the surrogate is HIGH (0.953 vs null mean 0.514) while RECIST falls below its null mean (0.434 vs 0.492) — the cleanest separation to date, confirming the Gide prognostication conflation is cohort-specific. Full results in Supplementary Table S20.

Fourth cancer type (ccRCC): IMmotion150 [23] (advanced renal cell carcinoma, atezolizumab arm; n = 165 RECIST-evaluable, 48 responders; iAtlas harmonization, which pools the monotherapy and combination arms). RECIST preserves two signals: GEP, the strongest fixed scorer in this cohort (0.662 [0.569–0.750], perm p = 4.4×10^{-4} , BH q = 0.002), and PD-L1 expression (0.599, p = 0.023, BH q = 0.045), the trial's own stratification biomarker. Switching RECIST to durable clinical benefit (PFS $\geq$ 6 months without progression) attenuates but does not abolish GEP (0.604, p = 0.016, BH q = 0.064). The time-only PFS < 6-month variant collapses every fixed signature (GEP 0.438) while ElasticNet-Var retains 0.659, consistent with trainable models fitting PFS-time correlates that are not ICI-specific. The surrogate endpoint yields GEP 0.905 (CDS 0.945 vs null 0.516). Full results in Supplementary Table S28; pipeline in scripts/e1_immotion150_executed.py.

Fifth cancer type (gastric): PRJEB25780 [24] (metastatic gastric cancer, n = 78, all anti-PD-1; TIGER portal). RECIST yields GEP 0.744 [0.602–0.854], perm p = 0.00034, BH q = 0.001 — the strongest GEP signal in the replication series and the first FDR-level confirmation in a non-melanoma, non-RCC tumor type. PD-L1 expression also survives (0.688 [0.537–0.820], perm p = 0.005, BH q = 0.005). CDS sits above its null (0.779 vs 0.601), confirming that the declared genes genuinely track response rather than creating a circular artifact. The surrogate endpoint inflates GEP to 0.922 (CDS 0.936 HIGH). Full results in Supplementary Table S29; pipeline in scripts/build_tiger_cohorts.py.

Clinical-clinical contrast (non-circular): One public cohort carries two clinical response definitions on the same patients: Liu 2019, whose iAtlas harmonization annotates both RECIST and PFS. We defined

durable clinical benefit (DCB) as progression-free at ≥ 6 months and excluded samples censored earlier. The two definitions agree substantially ($\kappa = 0.669$; 48 RECIST responders, 58 DCB). Across all six methods, switching between them moves predictions by at most 0.084 (ElasticNet-Var), roughly sixfold below the circular collapse (0.520 on Riaz) and below the largest within-cohort method gap on Liu RECIST (0.216, IMPRES vs TIDE). This is consistent with the circularity account, which predicts that neither clinical definition injects gene-label dependency into the feature space; as a single melanoma cohort with a complete-case construction, this contrast is supportive rather than dispositive. CDS places both endpoints in the genuine-biology band (0.721 and 0.726 against null means of 0.622 and 0.549). IMPRES is the only method permutation-significant under both definitions ($p = 0.016$ and 0.005), consistent with checkpoint-pair biology rather than a definition artifact. Full results in Supplementary Table S26.

Mechanism: The cytolytic score is operationally defined by GZMA and PRF1 expression^[14]. Learning methods predict it from co-expressed immune-infiltration genes, and GEP does so best (0.968) because GZMA/PRF1 are GEP members. RECIST — not gene-defined — removes that leverage (0.503, 0.465), while IMPRES, whose checkpoint-gene relations do not overlap the endpoint genes, stays near chance on both definitions. CDS flags the cytolytic endpoint HIGH (0.954) because endpoint-defining genes sit inside the feature space.

On the HGNC-mapped Riaz matrix, TIDE is near chance for both endpoints (RECIST 0.466, cytolytic 0.470), consistent with the tidepy implementation checked at source (`classification = correlation['TIDE'] < vthres`, default `vthres = 0`).

Design Choice 2: Feature Selection Determines Biological Interpretation

Within identical ElasticNet models on Hugo 2016 ($n = 28$, RECIST) we compared variance- and MI-based feature selection.

Table 2. ElasticNet on Hugo 2016 ($n = 28$): MI versus variance feature selection.

Metric	MI-based	Variance-based	Interpretation
AUROC (5-fold nested CV)	0.631	0.528	Not distinguishable (cross-seed $\Delta -0.008 \pm 0.050$, Table S12)
Gene overlap (top-500)	—	27.0% (135/500) prescreen convention; 2.4% (12/500) full-data MI	Overlap is convention-dependent (v33; two-stage prescreen is the pipeline's actual selection path — MI is computed within the top-2000 variance prescreen; the full-data MI variant is the alternative convention)
Spearman ρ (OOF scores)	—	0.75 ($p < 0.001$)	Highly correlated predictions
Observed Δ AUROC (MI – Var)	+0.103 (benchmark split)	-0.008 ± 0.050 (10 seeds, Table S12)	Includes zero
Permutation p (n=60 full pipeline)	0.131	0.344	Neither distinguishable from chance

Under the corrected, leakage-free implementation (v33) the two strategies achieve indistinguishable performance and highly correlated out-of-fold predictions (Spearman $\rho = 0.75$; Supplementary Figure S2); their top-500 gene sets overlap by 27% under the pipeline's two-stage prescreen convention (2.4% under

the full-data MI convention — the overlap's dependence on the convention is itself a design-choice effect; Supplementary Figure S3). Neither variant is distinguishable from chance by permutation ($p = 0.13$ and 0.34 at $n = 28$; Table 2). Feature selection at $n = 28$ therefore changes the selected gene sets — and the biological narrative — without measurably changing predictive performance (cross-seed stability: Supplementary Figure S7; Table S12). Learning curve analysis (Supplementary Figure S6; Table S13) shows the expected small-sample behavior: point estimates remain near chance at every subsample size while variance inflates steeply as n decreases (across-seed SD 0.211 at $n = 8$ vs. 0.038 at $n = 28$); the $n = 8$ point (~ 6 training samples per fold at $k = 5$) characterizes estimator variance rather than performance.

Design Choice 3: Endpoint Circularity Can Be Detected Before Benchmarking

Relation to Design Choice 1. The response-definition effect is mechanistically an instance of endpoint circularity: the cytolytic surrogate is operationally defined by genes inside the feature space. Design Choice 1 measured its *magnitude* within patients; Design Choice 3 asks whether circularity can be recognized *before* any benchmark is run, and develops the diagnostic. The corresponding non-circular contrast is measured directly: between the two clinical definitions on Liu 2019 (RECIST versus PFS-derived DCB), endpoint-switch deltas reach only 0.084 across six methods — the collapse is specific to gene-defined surrogates, not to clinical-definition disagreement (Results; Table S26).

Table 3. CDS with cohort-matched random-label nulls (200 replicates; declared genes GZMA/PRF1; v1.2.1, scores identical under v1.1.0 — the archived 10-replicate nulls give the same readings and remain in cds_nulls_v33.json). 'ML AUROC Range' = trainable methods only (ElasticNet MI/Var), v33 values. Riaz RECIST is reported on the HGNC matrix (0.409); the archived 0.685 was computed on the pre-v33 Entrez matrix.

Endpoint	CDS	null mean	null max	CDS vs null	ML AUROC Range	Inflated?
Riaz 2017 cytolytic	0.954	0.566	0.842	far above null max	0.968–0.985	Yes
Gide 2019 RECIST	0.852	0.555	0.800	above null max	0.605–0.630	No
Lauss 2017 RECIST (ACT)	0.781	0.578	0.866	above null mean	0.627–0.700	No
Riaz 2017 RECIST (v33)	0.409	0.560	0.836	within null	0.465–0.569	No
Hugo 2016 RECIST	0.449	0.586	0.834	within null	0.528–0.631	No

CDS places the only circular endpoint far above its random-label null (Riaz cytolytic 0.954 vs null mean 0.566 , max 0.842 ; Figure 2) — exactly where a GZMA/PRF1-defined endpoint must sit — with ML methods reaching 0.968 – 0.985 . The clinical endpoints sit at or inside their nulls (Hugo 0.449 vs 0.586 ; Riaz RECIST 0.409 vs 0.560 ; Lauss 0.781 vs 0.578 , max 0.866). Gide 2019 (Table S4b) returns 0.852 , above its null maximum (0.800): the declared genes genuinely track response there (GEP 0.830 , $q < 0.001$) — genuine biology that CDS v1.x cannot separate from circularity, so the categorical thresholds do not carry the validated claim. The validated claims are the null-referenced ordering above and the procedural rule that any gene-defined surrogate, of any program type (Supplementary Table S18), is circular by construction. The newly executed clinical replications return HIGH as well (GSE274975 RECIST 0.750 vs null mean 0.488 ; IMmotion150 RECIST 0.741 vs 0.492): in immune-inflamed tumors CDS does not separate circular from genuine endpoints, and the binding control is the declaration of endpoint-defining genes, with the continuous null-referenced score as a companion rather than a gate (Limitations). **Within-study consistency check:** CDS scores for Hugo, Riaz and Lauss preceded the corresponding ML benchmarks; all sat at or within their nulls (Table 3), and no trainable AUROC

exceeded 0.70 on them. The benchmarks therefore constitute a subsequent verification of the CDS readings — a within-study check, not prospective validation. External prospective validation remains open (Limitations).

External consistency check: The Auslander et al. (2018) study provides an independent validation scenario. IMPRES was trained on Auslander's data and validated externally on Hugo 2016 (RECIST), achieving AUROC 0.83 in the original paper (0.795 in our benchmark). Had CDS existed then, it would have returned 0.449 for Hugo RECIST (below its cohort-matched null mean of 0.586) — correctly indicating a non-circular endpoint and consistent with IMPRES reflecting checkpoint-pair biology rather than endpoint-gene dependency.

Complete Permutation Test Matrix

Figure 3 shows the complete 36-cell permutation matrix (observed AUROC and permutation p per method-cohort cell; asterisks mark within-cohort BH $q < 0.05$); the full numeric matrix, including the frozen-pipeline observed AUROC for every cell, is in Supplementary Table S11. PD-L1 cells use the fixed-scorer protocol (50,000 label shuffles against the fold-normalized CD274 score, which exactly reproduces the benchmark AUROC); the seven previously not-computed (n.c.) ElasticNet-MI/Var cells were completed with frozen hyperparameters and merged, so all 36 of 36 cells carry a recorded permutation p, with within-cohort BH families (six cells each) recomputed; XGBoost is excluded from this matrix and is reported separately in Methods. Permutation-run AUROCs deviate from the benchmark AUROC on the eight clinical trainable cells by 0.000–0.103 (frozen hyperparameters differ from the tune_primary selections; per-cell JSON under results/benchmark/v33/nc_cells/, extended 2026-09-18 in nc_cells_ext/). After within-cohort Benjamini-Hochberg correction (families of six cells per cohort-endpoint), the following survive. On Hugo 2016, IMPRES alone (0.795, $q = 0.017$; per-method concordance and leave-one-pair-out sensitivity in Supplementary Tables S5 and S6). On Gide 2019 ($n = 73$), GEP (0.830, $q < 0.001$), TIDE (0.752, $q < 0.001$), PD-L1 (0.791) and IMPRES ($q = 0.032$), and — after extension to 5,000 full-pipeline shuffles — the trainable signatures ElasticNet-MI (frozen-track 0.662, $q = 0.028$) and ElasticNet-Var (0.605, $q = 0.028$). On the circular Riaz cytolytic endpoint, ElasticNet (0.968, $q = 0.003$), ElasticNet-Var (0.985, $q = 0.002$), GEP and PD-L1 (0.868; both $q < 0.001$) and IMPRES ($q = 0.039$) — on a HIGH-circularity endpoint even methods whose features include the endpoint-defining genes survive, which is the circularity effect itself. The Lauss PD-L1 biomarker (0.780) is nominal-only after the completed six-test family correction ($p = 0.0096$, $q = 0.058$), and no other Lauss cell reaches $q < 0.05$ (the smallest other Lauss q is 0.072, ElasticNet-MI/Var at the extended 5,000-shuffle resolution). No trainable method survives FDR correction on any clinical endpoint at $n \leq 43$; on Gide ($n = 73$), ElasticNet-MI and ElasticNet-Var are detectable under the within-cohort family ($q = 0.028$, 0.028) though neither survives the global family (ElasticNet-Var $q = 0.066$; ElasticNet-MI $q = 0.070$; the re-run XGBoost cell discussed in Methods sits outside this reported matrix) — the first sample size at which any method's signal becomes detectable, consistent with the power analysis. IMPRES on Hugo carries the Carter et al. (2019) training-set-bias caveat [25]. Family-definition sensitivity (Supplementary Table S21): the Gide ElasticNet-Var, Gide IMPRES and Riaz-cytolytic IMPRES cells lose significance under global or method-class-stratified families, and the Lauss PD-L1 cell gains significance under them; the primary conclusions — no trainable-method FDR survival on clinical endpoints at $n \leq 43$, circular-endpoint inflation, IMPRES–Hugo and GEP/TIDE/PD-L1–Gide — hold under all three definitions. Cohort display

names match the Methods cohort table; internal dataset keys retain the legacy identifier Nathanson_2017 for machine readability (see Methods, Cohort Selection and Preprocessing).

Integrated Multi-Cohort Results

Across the four clinical-endpoint cohorts (Hugo, Lauss, Gide, Riaz RECIST; full AUROC/CI table in Supplementary Table S22, plotted in Supplementary Figure S8), the methods swap ranks: IMPRES is best on Hugo (0.795) and Riaz RECIST (0.613), GEP on Gide (0.830), and PD-L1 on Lauss (0.780), while neither ElasticNet variant leads any column. The between-method spread reaches 0.508 (Jung 2019: GEP 0.786 vs TIDE 0.278), so cross-cohort rankings conflate cohort effects with method effects — a spread that partly reflects deployment choices, since the Jung TIDE cell is computed under the tidepy default threshold convention (Methods) and sits at the two-sided edge of its permutation envelope (two-sided $p \approx 0.08$); either reading illustrates that cross-cohort rankings mix design, deployment and method effects. All trainable-method bootstrap CIs on clinical endpoints at $n \leq 43$ overlap 0.5, and no method ranking is claimed. On the fifth cohort-endpoint, Jung 2019 DCB ($n = 27$, NSCLC; corrected labels 1 = durable clinical benefit, 6/27; Supplementary Table S23), TIDE scores far below chance (0.278); with only 6 DCB responders this deviation sits inside the permutation null envelope ($p = 0.952$), and the value is reported as computed under the tidepy default threshold convention rather than re-tuned to the cohort.

Survival Analysis: AUROC Alone Does Not Capture Clinical Utility

On Hugo 2016, RECIST response strongly stratifies overall survival (OS; log-rank $p = 0.0003$ with true events — 26 patients / 12 deaths, with the local-versus-iAtlas metadata validation reported in Supplementary Table S24; Supplementary Figure S4), whereas IMPRES — the best AUROC method on this cohort — does not (KM $p = 0.554$). Cox regression with z-standardized signatures (HR per 1 SD) confirms the dissociation (IMPRES HR = 0.631, 95% CI [0.325–1.227], $p = 0.175$; GEP HR = 1.03, 95% CI [0.562–1.887], $p = 0.923$): AUROC for binary response prediction does not guarantee clinically meaningful survival stratification. Clinical-utility metrics for ElasticNet-Var on Hugo (v33 recomputed; full values in Supplementary Table S9) point the same way: decision-curve net benefit is positive but consistently below treat-all (0.345 vs 0.405 at $p_t = 0.10$; 0.259 vs 0.330 at 0.20; 0 at 0.50), and reclassification gains are absent (NRI = -0.113 , $p = 0.658$; IDI = -0.029). The argument is not an artifact of analyzing a weak method. IMPRES, the only FDR-significant method on this cohort (AUROC 0.795), fails the same way as a decision tool: at the 0.5 threshold it labels 92% of patients responders (sensitivity 0.923, specificity 0.267), its net benefit coincides with treat-all at the 10% and 20% thresholds, and its reclassification metrics do not exceed the random baseline (NRI = -0.20 , IDI = -0.073 ; Supplementary Table S9). Its score is an uncalibrated fraction, so these metrics reflect threshold behavior; either way, the ranking advantage of the cohort's only surviving method does not extend to decision utility. The local Lauss matrix contains no OS metadata, so the archived Lauss survival row could not be regenerated under the v33 protocol and is excluded (archived values retained in Supplementary Table S9 for continuity).

Uncertainty on the effect-size comparison. The $3.5\times$ is a descriptive ratio of benchmark point estimates ($\Delta = 0.520$ against a same-endpoint method gap of 0.147), not a bootstrap quantity. The inferential anchor is the Δ interval, which excludes zero under both hyperparameter constructions on the full 42-patient RECIST-evaluable set (frozen: $\Delta = 0.343$, bootstrap 95% CI 0.097–0.584; tuning: $\Delta = 0.386$, CI 0.163–0.598; 1,000 patient resamples, seed 42). The ratio itself is unstable because its denominator is small and volatile at this n (frozen-track ratio 11.98, CI 0.46–14.32; tuning-track constructs give ratios centered near

2.6, CI 0.52–3.18), so the 3.5× is reported as descriptive only and the existence claim rests on the Δ intervals. (results/benchmark/v33/effect_size_bootstrap_v33.json, tuning track in effect_size_bootstrap_tuning_v33.json)

Power Analysis: Minimum Sample Size Requirements

One-sample detectability. Table S25 quantifies two-method comparisons; one-sample statements (AUROC exceeding 0.5) are calibrated with the Hanley-McNeil single-sample variance (normal approximation, one-sided $\alpha = 0.05$; results/benchmark/v33/power_onesample_v33.json). At $n \leq 43$, 80% power requires a true AUROC of 0.74–0.79 (Riaz RECIST 0.75; Hugo 0.74; Lauss 0.76; Jung 0.79), whereas at $n = 73$ (Gide) it falls to 0.66: effects genuine at $n = 73$ in the 0.60–0.70 range sit below the detection floor at $n \leq 43$ — which is what the permutation matrix shows.

At $n = 25$ –43 the minimum detectable ΔAUROC is 0.21–0.23 (Supplementary Table S25; curves in Supplementary Figure S9). Pairwise comparison at ΔAUROC = 0.10 (80% power) requires ≈130 patients under the most favorable correlation assumption (paired, $r = 0.75$) and ≈260–510 under weaker or no correlation (power_table_v33.json); $n \geq 50$ is therefore a minimum practical floor, not a sufficient condition (Table 9); the floor is anchored to detectability — a true AUROC of 0.68 first becomes detectable at 80% power at $n \approx 55$ (40–55% responder fractions), and the smallest effect observed to survive FDR (ElasticNet-MI, 0.662 on Gide) requires $n \approx 71$ –79, a range that brackets the Gide cohort size (73). **Table 9. Minimum requirements for reliable immunotherapy prediction benchmarking.**

Requirement	Current Practice	Recommendation
Patients per cohort	$n = 25$ –43	$n \geq 50$ minimum floor; ≈130 for pairwise ΔAUROC = 0.10 at 80% power (paired, $r = 0.75$; ≈260–510 under weaker correlation)
Independent cohorts	1–2	≥3 with consistent endpoint definitions
Response definitions	Mixed (RECIST, PFS, surrogate)	≥1 clinical endpoint; surrogates labeled with defining genes
Feature selection	Often unreported	≥2 strategies pre-specified; report overlap and cross-seed stability
Circularity check	Not performed	Compute CDS; report continuous value; calibrate thresholds on synthetic data
Non-learning reference baselines	Often omitted	Include non-learning baselines as negative controls (IMPRES, which does not overlap endpoint genes; GEP's own collapse instead demonstrates the mechanism, since its genes contain GZMA/PRF1)
Permutation test	Rarely performed	Full permutation matrix; FDR correction; ≥5000 permutations for borderline cases†
Statistical reporting	AUROC only; $p < 0.05$	Exact p-values; 95% CIs; effect sizes; cross-fold SD at $n < 50$
Reproducibility	Variable	Docker container + Zenodo archived data

†Trainable-method cells use 2,000–5,000 full-pipeline shuffles (each shuffle re-fits the pipeline, ≈60–90 s; all eight clinical trainable cells extended in 2026-09-18, four cells to 5,000); ≥50,000 applies to fixed scorers, and the resolution-hardening protocol is described in Methods. **Priority guidance:** The three largest-variance levers are (1) at least one clinical endpoint alongside any molecular surrogate, (2) pre-specified multi-strategy feature selection with overlap reporting, and (3) non-learning reference baselines, with IMPRES as the negative control and GEP's own collapse as the mechanism demonstration; the remaining recommendations add rigor but address smaller-magnitude variation.

DISCUSSION

A predictive account of when gene-defined surrogates mislead

The three modules compose into one statement: a gene-defined surrogate misleads in proportion to (i) the immune-infiltration-driven co-expression supply linking endpoint genes to the feature space (high in melanoma, low in COAD-like contexts — the TCGA gradient below), (ii) the overlap between declared endpoint genes and the feature space (complete for the cytolytic surrogate, absent for IMPRES), and (iii) sample size, which sets the detectability floor (nothing below $n \approx 43$) and the power to separate genuine signal from construction (single methods from $n \approx 73$, two-method comparisons from $n \approx 130$; Table S25). These anchors form a testable claim — in a low-infiltration tumor type the endpoint-switch collapse should shrink toward the method gap even at adequate n — and the 36-cell matrix and CDS nulls make it checkable. The TCGA anchor is prognostic and its SKCM–COAD contrast partially conflates cancer type with metastatic-versus-primary composition (the SKCM signal is metastatic-carried; Table S15), so it conditions rather than demonstrates the account. The account also predicts that two clinical definitions of the same construct should barely move predictions; the Liu 2019 RECIST-versus-DCB contrast provides direct support for this (largest endpoint-switch $\Delta = 0.084$ across six methods, below the largest within-cohort method gap; Table S26), while remaining a single-cohort, complete-case comparison. SPATBench asks a different question from existing benchmarks^[15–18], which hold methodology fixed: which design choices cause that unreliability at available sample sizes, and does the answer convert into design rules? The answer: response definition exerts a larger effect (largest within-cohort $\Delta = 0.520$, ElasticNet-Var) than any between-method difference on the same RECIST endpoint (max $\Delta = 0.147$); across cohorts the between-method spread reaches 0.508 (Jung), which is why the endpoint effect must be demonstrated within-cohort — at $n = 43$, *how you define response matters more than which algorithm you choose*.

The mechanism ports across three further cohorts with executed endpoint-switch replications (Liu 2019, melanoma; GSE274975, NSCLC; IMmotion150, renal cell carcinoma; STAD_PRJEB25780, gastric cancer), alongside the IMvigor210 cytolytic-to-RECIST switch on 297 urothelial-carcinoma patients (largest $\Delta = 0.354$ against a same-cohort method gap of 0.180, CDS 0.953 versus 0.434), plus the Gide surrogate-construction control (CDS 0.955); all use the same author-constructed cytolytic surrogate, so these establish cross-cancer portability of the circular construction on clinical endpoints, not that clinical RECIST rankings differ across every tumor type (five cancer types now carry executed external replications; Supplementary Tables S19, S20, S27, S28 and S29); the published-study audit (Tables S17, S20) establishes cross-study consistency of the reading on the computable cohorts.

Of the Table 9 rules, the genuinely new ones from this study are requiring a clinical endpoint alongside any gene-defined surrogate, declaring endpoint-defining genes, and computing CDS; the remaining rows reinforce standard reproducibility hygiene. Beyond immunotherapy, any transcriptomic prediction task using a gene-defined surrogate carries the same circularity risk, and CDS generalizes to any expression matrix paired with a binary label. We use "design choice" rather than "confounder" throughout: the three variables are under experimental control — *variance-dominating factors in small-sample benchmarks* — and should therefore be controlled by experimental design, not merely adjusted for statistically.

The Response Definition Problem

The cytolytic $\rightarrow$ RECIST collapse (largest $\Delta = 0.520$, ElasticNet-Var) generalizes: any molecular surrogate operationally defined by genes in the feature space creates a circular prediction task. The corresponding rules are distilled in Table 9 (clinical endpoint required; surrogates labeled with their defining genes; CDS computed; non-learning baselines included); one rule is not captured there: for endpoints with CDS > 0.80 , a conservative operational HIGH threshold, assume ML methods are learning the endpoint-defining genes until independent clinical-endpoint validation proves otherwise; genuine endpoints in Table 3 reach null maxima of 0.80–0.87, so this rule deliberately trades false alarms for safety.

Feature Selection as a Narrative Design Choice

The substantial overlap of MI- and variance-selected gene sets (27% of the top-500) with highly correlated out-of-fold predictions ($\rho = 0.75$) shows that feature selection determines the biological story even when AUROC is statistically indistinguishable at $n = 28$. The reporting rules are in Table 9 (≥ 2 strategies pre-specified; overlap reported; cross-seed stability at $n < 50$); the one addition: accompany biological interpretation with sensitivity analysis — overlap reporting is the single most impactful check, directly revealing whether conclusions are strategy-dependent.

On Method Context and Generalizability

Our data are consistent with method performance being inherently context-dependent ^[15,17]: IMPRES's AUROC ranges from 0.795 (Hugo) to 0.583 (Jung DCB), and GEP's from 0.446 (Hugo) to 0.830 (Gide). IMPRES's context-sensitivity is consistent with Carter et al. (2019), who showed IMPRES failing in an independent melanoma cohort (AUROC 0.52–0.55) ^[25]; the inherited feature-selection-bias caveat is in Limitations. Benchmark studies should therefore report each method's known developmental limitations alongside performance. We note that benchmark unreliability does not imply that a biomarker is clinically useless — a predictor with wide CIs at $n = 28$ may still capture genuine biology that a larger cohort would confirm. The point is narrower: at current sample sizes, one cannot *distinguish* reliable from unreliable benchmarks by AUROC alone, and the design choices documented here are the reason. The GEP's clinical development (as part of the approved Vantage assay) proceeded despite the same small-cohort limitations documented here — trial design, not retrospective AUROC benchmarking, was the arbiter of utility.

Independent Prognostic Evidence: TCGA Cancer-Type Gradient

The TCGA analysis is **prognostic, not predictive** — univariate Cox models of overall survival without clinical covariates (stage, age) — and cannot validate treatment-response prediction; it provides large-scale independent evidence that immune-signature behavior reflects method–context interaction. **v33 correction (data-provenance audit)**. Re-running the pipeline against the repository's current data reproduced three of the four archived cohorts and revealed two provenance defects, disclosed here rather than silently re-used. First, SKCM sample-type policy: the archived analysis used all tumor samples ($n = 426$ of 444; 367 metastatic, 76 primary); the restored all-tumor policy yields $n = 441$ patients (212 deaths), with **11/12 immune signatures significant after Benjamini-Hochberg correction** (HR 0.795–0.928; BH $p \leq 2.7e-3$) — the *stromal* signature is the only non-significant one and the only one with HR > 1 (1.086, BH $p = 0.090$), an internal negative control consistent with immune-specific rather than generic prognostic biology; the signal is also not a stage- or age-proxy — in multivariate Cox models adjusted for age and

AJCC stage, 8/12 signatures remain BH-significant with stronger protective HRs (0.569–0.704; Supplementary Table S15). Second, the local COAD expression file was not TCGA-formatted and could not support the archived COAD result (n = 588, 119 deaths), so v33 withdrew it and re-acquired the cohort from the GDC current index (STAR counts, GENCODE v36, primary tumors; scripts/build_coad_gdc.py), yielding n = 458 patients, 102 deaths, with **0/12 signatures significant (all HR $\approx$ 1)** — independently reproducing the archived finding (the GDC STAR source differs from the other three cancers' cBioPortal RSEM — disclosed; it does not affect within-cancer statistics). The original report used nominal $p < 0.05$ only; BH-adjusted values within each cancer type strengthen the SKCM conclusion by surviving FDR correction at 48 tests. The TCGA validation comprises 2,491 patients (SKCM 441 + BRCA 1,082 + LUAD 510 + COAD 458; 12 signatures $\times$ 4 cancer types; data/tcga/tcga_cox_all.json). A pronounced cancer-type gradient emerges (Supplementary Figure S11; Supplementary Table S15; BH within each cancer type): 11/12 signatures significant in SKCM, 3/12 in BRCA (CD8 T cells, B cells, TLS), 1/12 in LUAD (B cells) and 0/12 in COAD — an ordered decrease (intermediate steps rest on one to three signatures each; descriptive, not a calibrated dose–response). The B-cell signature is significant in SKCM, BRCA and LUAD but not COAD (HR 0.904/0.885/0.873 vs 1.153); CD8 T-cell and TLS are significant in SKCM and BRCA only; IFNG-response only in SKCM. Immune signatures point protectively in SKCM/BRCA/LUAD while COAD HRs hover at or slightly above 1 (e.g. CD8 T cells 1.177) — COAD reproduces the archived finding that immune infiltration is not prognostic there. No time-dependent AUC is reported under v33 (archived 0.33–0.60 not regenerated). This pattern matches Usset et al. (2024), who showed that thousands of published biomarkers collapse into five latent factors [26]. Full methods in Supplementary Note S4.

CDS: Current Status and Validation Roadmap

CDS is itself subject to the same validation standards we recommend for benchmarking tools — an epistemological tension we acknowledge. Its current status:

- **What CDS can do:** Quantify endpoint–gene association on a null-referenced scale: the circular Riaz cytolytic endpoint sits far above its null (0.954 vs null mean 0.566, max 0.842) while the clinical endpoints sit at or within their nulls (Hugo RECIST 0.449, Riaz RECIST 0.409; Table 3). Provide a lightweight (<10 s) diagnostic requiring only an expression matrix, binary labels and declared endpoint genes (pip installable; NumPy/SciPy/scikit-learn only). Support the procedural circularity rule — any endpoint defined by genes in the feature space is circular by construction, whatever its score — shown program-agnostic by the negative-control stress test (Table S18). Retrospective application to the Auslander et al. (2018) external validation of IMPRES on Hugo is consistent with genuine signal rather than circular inflation (Results) [5,27]. The same reading extends to the published benchmark studies whose cohorts are computable — Auslander et al. on Hugo, and Jiang/Mariathasan via IMvigor210 (Table S17a–b): in every computable cohort the cytolytic positive controls sit far above their nulls while the clinical endpoints sit at or within theirs (with the notable exception of STAD_PRJEB25780 gastric cancer, where RECIST CDS 0.779 sits above its null mean of 0.601, consistent with the declared genes genuinely tracking response) (an independent 10-replicate null run preserves the ordering — Tables S17/S19/S20); the Chowell and Liang multi-cohort designs remain unverified, and the Van Allen CTLA-4 cohort [28] remains expression-inaccessible. (The Gide 2019 value in Table S4b is a v33 recomputation of a score never stored at analysis time.)

- **Practical pre-benchmarking workflow:** (1) Declare which genes, if any, operationally define the endpoint; a gene-defined surrogate is circular by construction — switch to a clinical endpoint or accept that ML AUROC will be inflated by endpoint–gene co-expression. (2) Compute CDS on the declared genes before training any model. (3) Report the continuous CDS value against a 200-replicate random-label null on the same cohort and feature space; a value far above the null maximum flags strong gene–label association — circular construction or genuine biology, distinguishable only by provenance. (4) Treat the shipped categorical labels (HIGH > 0.70, MODERATE 0.30–0.70, LOW ≤ 0.30) as advisory: Table S18 shows they carry no specificity among structured endpoints. No ML training is required.
- **What CDS cannot yet do:** Prospectively validate on independent multi-endpoint cohorts (none is publicly available with the required dual-endpoint design). Provide well-calibrated categorical thresholds (provisional; recalibrated from synthetic data with $n = 30$ null replicates). Distinguish among MODERATE-risk endpoints (C3 saturation limits discrimination in the 0.50–0.80 range). Be computed on cohorts where the raw expression data is not publicly accessible. **CDS: Known Failure Modes.** (1) **Undeclared endpoint genes (false negatives):** undisclosed molecular surrogates evade detection; if CDS is LOW but ML AUROC > 0.80, suspect undeclared endpoint–gene dependency. CDS v2.0 will add a hypothesis-free mode that permutation-tests the full feature-label MI distribution. (2) **Label noise (false negatives):** label misclassification attenuates gene-label correlations and may return LOW risk; compare CDS between the suspect labels and a gold-standard endpoint on the same cohort — a large discrepancy suggests label-quality issues. (3) **C3 saturation in immune-dominant contexts (loss of discrimination):** in highly immune-infiltrated cancers (melanoma, NSCLC), C3 saturates at 1.000 on every real cohort and all 60 pseudo-endpoints (Table S18), leaving C1 and C2 as the sole discrimination sources; in lower-infiltration tumors (UC, gastric), CDS on clinical endpoints remains at or below null, confirming that saturation is infiltration-dependent rather than universal; CDS v2.0 will use a permutation-based C3 to estimate the null distribution of feature-space enrichment.
- **Validation roadmap:** (1) Re-calibrate C3 using pathway-stratified feature selection (CDS v2.0). (2) Validate on external multi-endpoint cohorts as they become available (executed on Liu 2019, GSE274975, IMmotion150 and STAD_PRJEB25780; the remaining pre-screened candidate, NIVACTOR, is microarray-based). (3) Third-party replication on independent benchmark endpoints before full evaluations. (4) Pre-registration of CDS scores, enabling prospective validation against subsequently observed AUROC values.

Limitations

1. **Sample size:** All 95% CIs for the trainable methods on clinical endpoints at $n \leq 43$ overlap 0.5 (the sole fixed-scorer exception at $n \leq 43$ is IMPRES on Hugo — the headline positive result, which does not bear on method ranking among trained models; Lauss PD-L1 is nominal-only, $q = 0.058$). IMPRES's CI width (0.350 on Hugo) quantifies the genuine uncertainty at $n = 28$. No ML method is statistically superior for $n \leq 43$.
2. **Response-definition evidence base:** The cytolytic→RECIST collapse was demonstrated on Riaz 2017, replicated on five independent cohorts — Liu 2019 (melanoma, $n = 122$; Table S19), IMvigor210 (bladder carcinoma, $n = 297$; Table S20), GSE274975 (NSCLC, $n = 61$; Table S27), IMmotion150 (ccRCC, $n = 165$; Table S28) and STAD_PRJEB25780 (gastric cancer, $n = 78$; Table

S29) — using the same author-constructed surrogate, and is k-fold robust (Supplementary Table S16). The collapse magnitude (largest $\Delta = 0.520$) may be cohort-specific: where immune infiltration is weaker (in TCGA, 0–1 of 12 signatures reach FDR-adjusted significance in LUAD and none in COAD), the effect should attenuate because less immune co-expression is available to leverage. The clinical–clinical contrast bounds the non-circular component: RECIST versus PFS-derived DCB on Liu 2019 moves no method by more than 0.084 (Table S26), so the demonstrated collapse is attributable to the circularity channel rather than to clinical-definition disagreement. The executed external evidence spans five cancer types, and the remaining boundary is cohort availability rather than design. Two data realities constrain scope: GSE274975 keeps its clinical endpoints in the source publication's supplementary table rather than in any repository metadata field, and IMmotion150's harmonization pools the atezolizumab monotherapy and combination arms. The remaining pre-screened candidate (NIVACTOR, HNSCC) is microarray-based with 12 responders and was set aside pending a per-patient response table outside PDF supplementary material. The IMvigor210 replication ($n = 297$, bladder carcinoma) already demonstrates that the framework generalizes beyond melanoma when the data exist.

3. **CDS calibration, program-agnosticity and C3 formulation:** The thresholds are provisional, and the negative-control stress test (Supplementary Table S18) shows the labels have no specificity among structured endpoints: all 60 pseudo-endpoints score HIGH (0.79–0.95), and 20–36% of 200 random-label replicates exceed the shipped 0.70 boundary on every real cohort (`cds_nulls_200_v33.json`) versus $\approx 3\%$ on the synthetic calibration null — the shipped labels overstate specificity on real data by an order of magnitude. CDS's validated content is the continuous value against a cohort-matched random-label null; circularity protection is procedural (declare the endpoint-defining genes). The Gide 2019 RECIST value (0.852 vs null max 0.800; GEP 0.830 marks the genuine biology) quantifies the core limitation: CDS v1.x cannot separate genuine biology from circularity. C3 saturates at 1.000 on all real cohorts because highly variable genes in bulk tumor RNA-seq are predominantly immune-related; the composite is bounded to $[0.40, 1.00]$ there, making the shipped LOW tier (≤ 0.30) mathematically unreachable. With C3 saturated, CDS on real bulk data reduces to an affine transform of C1 and C2 ($0.4 + 0.3 \cdot (C1 + C2)$); the constant offset shifts the real-cohort distribution upward relative to the synthetic calibration — which is why the categorical tiers do not transfer to real cohorts (20–36% of real random-label replicates exceed 0.70 versus $\approx 3\%$ on the synthetic null) and why the cohort-matched null, which absorbs the offset, is the validated reading.
4. **Data access and cross-cohort validation:** IMPRES, GEP and TIDE require HGNC-mapped features; the originally archived Jung 2019 matrix carried Ensembl identifiers without such mapping, and the HGNC-mapped matrix now used for all methods restores their evaluation. The Jung label orientation was corrected (the release codes 1 = no durable benefit — the inverse of every other cohort; all analyses use 1 = DCB). Leave-one-cohort-out validation remains precluded by gene-identifier heterogeneity across cohorts — a barrier that is itself a finding; CDS's data-access requirement (public expression matrix + labels) serves as a de facto transparency enforcement mechanism.
5. **Clinical biomarker proxies:** TMB and PD-L1 IHC results use gene expression proxies (constant baseline and CD274 expression, respectively).
6. **Cohort composition:** The immunotherapy cohorts are predominantly melanoma (169/196 patients), and one of the five is adoptive T-cell therapy rather than ICI (Methods). The TCGA analysis provides

cross-cancer prognostic evidence (2,491 patients, 4 cancer types). The fully controlled within-patient endpoint-switch contrast is Riaz melanoma, while the surrogate-versus-clinical inflation pattern is observed in all five executed cancer types; CDS calibration rests on the benchmark cohorts and their nulls. Gide 2019 includes both monotherapy and combination-therapy arms, introducing treatment heterogeneity.

7. **Learning curve caveat:** The $n = 8$ subsample analysis ($k = 5$, ~6 training samples per fold) is diagnostic rather than a precise estimate.

8. **IMPRES interpretative caveat:** IMPRES's Hugo performance (0.795, $q = 0.017$) may partially reflect inherited feature-selection bias from the original training protocol (34% non-random repeats) [25]; the balance between genuine checkpoint biology and inherited bias is unresolved.

STAR METHODS

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Zhixian Huang (huangzhixian324@gmail.com), the sole author of this study.

Materials availability

This study did not generate new materials.

Data and code availability

- Data: All raw cohort data are publicly available from GEO (GSE78220, GSE91061, GSE100797, GSE135222, GSE274975) and cBioPortal (mel_iatlas_gide_2019, blca_iatlas_imvigor210_2017, mel_iatlas_liu_2019, rcc_iatlas_immotion150_2018) and ENA (PRJEB25780). Preprocessed expression matrices, all per-cell permutation outputs (results/benchmark/v33/nc_cells/ and nc_cells_ext/) and figures are archived in the versioned Zenodo record (DOI: 10.5281/zenodo.22746254).
- Code: All original code is publicly available. SPATBench pipeline (v33): <https://github.com/huangzhixian324-dev/spa-bench> (MIT). CDS package (v1.2.1): <https://github.com/huangzhixian324-dev/cds> (MIT); the packaged source is also included in the Zenodo archive (cds_tool/, installable with `pip install ./cds_tool`). A versioned archive with a DOI is deposited on Zenodo (DOI: 10.5281/zenodo.22746254). The included Dockerfile builds the image (`docker build -t spa-bench:v33 .`) that reproduces the synthetic validation test (`scripts/validate.py`, exit code 0 verified) and ships the complete pipeline code; reproducing the reported tables additionally requires the Zenodo-archived inputs and `scripts/rerun_v33.py`.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

This study is purely computational and generated no new experimental data: no experimental models, human participants, or animals were used. All cohorts are publicly available, de-identified bulk RNA-seq datasets from published studies (GEO accessions GSE78220, GSE91061, GSE100797, GSE135222 and GSE274975; cBioPortal studies mel_iatlas_gide_2019, mel_iatlas_liu_2019, blca_iatlas_invigor210_2017 and rcc_iatlas_immotion150_2018; TCGA via the GDC), analyzed under the terms of their original publications.

METHOD DETAILS

Cohort Selection and Preprocessing

Five publicly available cohorts with pre-treatment bulk RNA-seq data and annotated clinical outcomes were analyzed (Supplementary Table S3):

Cohort	Accession	Cancer	n	Treatment	Endpoint	Responders
Hugo 2016	GSE78220 [8]	Melanoma	28	Pembrolizumab	RECIST v1.1	13 (46%)
Riaz 2017	GSE91061 [7]	Melanoma	43 (42 with RECIST)	Nivolumab	RECIST + Cytolytic	9 RECIST (of 42) / 21 Cytolytic (of 43)
Lauss 2017 (ACT)	GSE100797 [29]	Melanoma	25	Adoptive TIL	RECIST v1.1	10 (40%)
Jung 2019	GSE135222 [30]	NSCLC	27	Anti-PD-1/PD-L1	DCB vs NDB (RECIST v1.1, ≥6-mo follow-up)	6 DCB (22%)
Gide 2019	mel_iatlas_gide_2019 [31]	Melanoma	73	Anti-PD-1 ± anti-CTLA-4	RECIST v1.1	40 (55%)

Total: 196 patients (28 + 43 + 25 + 27 + 73). Cohorts were selected based on availability of pre-treatment bulk RNA-seq data with public clinical annotations. One cohort is adoptive T-cell therapy rather than checkpoint blockade: GSE100797 is the cohort of Lauss et al. (2017) [29] — referred to as "Nathanson" in parts of the benchmarking literature. The archived dataset key Nathanson_2017 is a legacy machine-readable alias retained for row-level reproducibility of the published tables, not an attribution. It is included deliberately as a stress test of method context-dependence beyond ICI; no headline claim depends on it, and we refer to the five cohorts as immunotherapy-treated throughout. The Riaz 2017 cohort provides two response definitions on the same patients — RECIST v1.1 from GEO SOFT metadata (of 49 pre-treatment biopsies with RECIST annotations, 9 had PR/CR and 33 had PD/SD; 7 had unknown response and were excluded, leaving **n** = 42 for the RECIST endpoint; the cytolytic endpoint uses all **n** = 43) and cytolytic activity (median-split of geometric mean of GZMA and PRF1; 21 above median [14]) — enabling a within-cohort, within-patient comparison of response definition effects. Cohorts were not pooled for meta-analysis due to endpoint heterogeneity (RECIST, DCB/PFS, and molecular surrogates capture different biology).

Label semantics: Throughout this work label 1 = clinical benefit (RECIST responder / DCB). For Jung 2019 (GSE135222), the original release coded 1 = NDB — the inverse of every other cohort; all Jung analyses use corrected labels (1 = DCB, 6/27).

All five benchmark cohorts' expression data were standardized to $\log_2(\text{FPKM} + 1)$; the executed external replications use cohort-appropriate normalization ($\log_2(\text{CPM} + 1)$) for the GSE274975 raw counts;

log2(TPM + 1) for the IMmotion150 iAtlas profile; Supplementary Tables S27 and S28). Patient-stratified k-fold cross-validation used k = 5 for all five cohorts except the Riaz RECIST endpoint, where the small responder count (9 of 42) constrained the design to k = 2. To assess k-fold robustness for the Riaz RECIST endpoint, we repeated the analysis with k = 3 and k = 5 (10 stratified random seeds each) and confirmed that the cytolytic→RECIST collapse magnitude is not an artifact of the k-fold choice (Supplementary Table S16; regenerated values in results/benchmark/v33/riaz_recist_kfold_v33.json). Gene symbols were mapped to HGNC where possible (Riaz 2017: Entrez-to-HGNC, 21,769/22,187 genes; Jung 2019: Ensembl-to-HGNC).

Methods Evaluated

Gene signatures (no training required): GEP (the 8-gene Ayers IFN- γ /T-effector signature: CD8A, GZMA, GZMB, IFNG, CXCL9, CXCL10, PRF1, TBX21 [3]; the circularity channel analyzed in this study is specific to gene sets containing GZMA/PRF1 — the 18-gene expanded GEP does not contain them and would not be expected to show the same collapse), TIDE (tidepy v1.3.9; the implementation was checked at source — classification = correlation['TIDE'] < vthres, default vthres = 0 — consistent with its near-chance scores on both Riaz endpoints, Table 1), IMPRES (the 15 canonical pairwise checkpoint-gene relations of Auslander et al. [5]; the relations were decoded from the authors' published feature set (github.com/noamaus/IMPRES-codes, FEATS.mat) and cross-checked against the 9 pairs named in the paper text (9/9 match); the remaining 6 relations are not independently verified beyond the authors' released file. For a sample, the IMPRES score is the fraction of the 15 logical relations (e.g., CD40 > PD-1) whose expression ordering holds, linearly rescaled when features are missing, exactly as described in the original Methods. No retraining is performed, mirroring the original design in which features were selected once on neuroblastoma spontaneous-regression data.)

Gene expression biomarker: PD-L1 (CD274 expression; proxy for IHC — real IHC scores unavailable in public metadata).

Machine learning (training required): ElasticNet logistic regression (sklearn saga solver, penalty = 'elasticnet') with variance-based feature selection (top-500 genes by expression variance in the training fold), ElasticNet with mutual information-based feature selection (top-500 genes by MI with the response label in the training fold), and XGBoost gradient boosting. ElasticNet hyperparameters (C, l1_ratio) were tuned via 3-fold inner cross-validation within each training fold; the selected hyperparameters were frozen during permutation testing, as is standard.

Supplementary baselines: TMB (constant 0.5; real mutation data unavailable) [32]. Immune deconvolution signatures (14 cell types, 43 unique genes; Supplementary Table S10) [33].

The six benchmarked prediction methods span four categories. All feature selection used only training-fold data.

Evaluation Protocol

Patient-stratified k-fold cross-validation with identical splits across all methods within each cohort. Ten metrics were computed. Here we report AUROC as the primary metric with 95% bootstrap confidence intervals (1,000 iterations) [27]. Expression-scale, preprocessing and cross-validation-strategy sensitivity analyses on Hugo 2016 (ElasticNet-MI) are provided in Supplementary Figure S5 and Tables S7 and S8.

AUROC, AUPRC and bootstrap CIs for every method–cohort cell are provided in Supplementary Table S9, with clinical-utility metrics (Brier score, DCA, NRI, IDI) for a representative trainable cell; per-fold values are fully determined by the archived split files (Supplementary Table S1). For cohorts with $n \leq 28$, we additionally report cross-fold standard deviation (SD) as a stability metric.

Permutation Testing

Full permutation tests were performed for all trainable method–cohort pairs. Exact p-values = (number of permuted AUROCs $\geq$ observed + 1) / (n_permutations + 1). Benjamini-Hochberg FDR correction was applied within each cohort-endpoint family (six methods per family; sensitivity to alternative family definitions in Supplementary Table S21) [34].

Inclusion logic: Two protocols are used, matched to each method class. (a) Fixed scorers — IMPRES, GEP, TIDE, and PD-L1 (CD274 expression) — produce scores that are independent of the labels, so the full-pipeline label-shuffle null is mathematically identical to shuffling labels against the fixed out-of-fold scores; each cell therefore uses 50,000 prediction shuffles (seconds of compute, exact to 1/50,001). This equivalence was verified by running both protocols on Hugo IMPRES. (b) Trainable methods — ElasticNet-MI and ElasticNet-Var — are re-fit end-to-end under every label shuffle (per-fold feature selection with frozen hyperparameters), with a per-cell number of shuffles recorded in Supplementary Table S11. XGBoost is excluded from benchmark reporting (degenerate on all clinical endpoints; its only non-degenerate AUROC sits on the circular cytolytic endpoint — development-phase evaluation only). Re-examination (v33 protocol): under the current protocol XGBoost is not degenerate — re-runs produce non-degenerate predictions (Hugo 0.554, Lauss 0.493, Jung 0.611, Riaz-RECIST 0.434, Gide 0.718, and 0.991 on the circular cytolytic endpoint; all six cohorts re-run; results/benchmark/v33/xgb_exclusion_evidence_v33.json); it shows no signal on any clinical endpoint at $n \leq 43$, so the negative claim is unaffected. A 1,000-shuffle full-pipeline permutation test of the Gide re-run (frozen hyperparameters, the Table S11 protocol) gives $p = 0.004$ and $q = 0.007$ in the within-cohort family recomputed over seven cells, with the six original Gide q-values unchanged or more conservative: XGBoost is the third detectable trainable signature at $n = 73$, and the detectability-onset reading now covers all three trainable methods tested. XGBoost remains outside the reported 36-cell matrix because the development-phase runs were not retained under the current protocol; the re-run and permutation evidence files are retained for audit (permutation record: results/benchmark/v33/nc_cells/Gide_2019_cBio_XGBoost.json; xgboost 3.4.1) even though the prediction tables are not.

Permutation resolution: p-values are exact given the recorded per-cell shuffle count: 50,000 for fixed scorers (minimum detectable $p = 2 \times 10^{-5}$) and 2,000–5,000 full-pipeline shuffles for the trainable methods, where the compute cost of re-fitting under every shuffle (≈ 60 – 90 s per shuffle) is the binding constraint. The initial budget ranged from 60 to 1,000 shuffles, under which the smallest achievable exact p at 60 shuffles ($1/61 \approx 0.016$) would have imposed a resolution ceiling inside a six-test cohort family. All eight clinical trainable cells were therefore extended to a uniform minimum of 2,000 shuffles (Hugo and Lauss MI/Var to 5,000; Jung and Riaz-RECIST MI/Var to 2,000) in a pre-registered resolution-hardening run (scripts/extend_nc_cells_5000.py, merged by scripts/merge_nc_cells_ext.py); no cell changed its significance verdict. The one borderline trainable cell, ElasticNet-MI on Gide 2019 ($n = 73$), had earlier been extended to 5,000 full-pipeline shuffles (checkpointed resumable protocol, frozen hyperparameters): the observed AUROC under this protocol is 0.662 (0.033 above the benchmark 0.629 — frozen-versus-

tuned hyperparameters, the same deviation class disclosed in the Supplementary Table S11 footnote), giving $p = 0.0232$ and within-cohort BH $q = 0.028$ — **FDR-significant**. ElasticNet-MI therefore joins ElasticNet-Var as a detectable trainable signature on Gide ($n = 73$), strengthening the reading that Gide is the smallest sample size at which trainable transcriptomic signatures become detectable on a clinical endpoint; our headline claim is unchanged, since it is scoped to $n \leq 43$, where — at the extended resolution — no trainable method survives (the smallest within-cohort trainable q at $n \leq 43$ is 0.0716, Lauss ElasticNet-MI/Var; the smallest q on any cell is 0.058, Lauss PD-L1), independently corroborated by bootstrap 95% CIs that all overlap 0.5 (Supplementary Table S9).

Multiple Comparison Framework

Three levels of inference: (i) Exploratory: AUROC point estimates with 95% bootstrap CIs reported without multiplicity correction (descriptive quantities). (ii) Permutation testing: all evaluated method-cohort pairs with within-cohort FDR correction. (iii) Pairwise comparisons of AUROCs via paired bootstrap tests of Δ AUROC. The analytic DeLong covariance method ^[35] was considered but the bootstrap version was used throughout — we refer to these as paired bootstrap tests, not DeLong tests). The Bonferroni threshold ($p < 0.05/36 \approx 0.0014$) is unattainable at current sample sizes — we treat this inaccessibility as a finding. Because family definition is an analysis degree-of-freedom, Supplementary Table S21 reports the full 36-cell matrix under three family definitions (within-cohort 6×6 ; global 36; method-class-stratified $24+12$): the primary conclusions are family-robust, and the four family-dependent cells are disclosed there.

Effect-Size Comparison Framework

To compare the relative influence of methodological choices against method choice, we computed the range of AUROC values attributable to each source on the Riaz 2017 cohort: (a) between-endpoint Δ AUROC for each method; (b) between-method Δ AUROC within each endpoint; (c) between-feature-selection Δ AUROC within ElasticNet. We report these as effect-size comparisons rather than variance-component percentages, which are unreliable with limited independent data points. Between-method spreads differ across cohorts, so the endpoint effect must be compared with method effects at the same level of control. Within the same patients and the same RECIST endpoint (Riaz), the largest between-method Δ AUROC is 0.147 (IMPRES 0.613 vs TIDE 0.466), so the largest between-endpoint effect (Var: $0.985 \rightarrow 0.465$, $\Delta = 0.520$) exceeds it by $3.5\times$; the same comparison yields $\approx 2.2\times$ on Liu 2019 ($0.425/0.195$) and $\approx 2.0\times$ on IMvigor210 ($0.354/0.180$) — the direction replicates, with a smaller ratio. The effect is further supported by non-overlapping bootstrap confidence intervals. The cytolytic-endpoint AUROC CIs for ElasticNet-Var ($[0.954, 1.000]$) and ElasticNet-MI ($[0.909, 1.000]$) do not overlap the RECIST-endpoint CIs ($[0.208, 0.704]$ and $[0.370, 0.755]$ respectively), and the same non-overlap pattern holds for the external replications (Liu 2019: cytolytic GEP $[0.928, 0.986]$ vs RECIST $[0.427, 0.642]$; IMvigor210: cytolytic $[0.948, 0.981]$ vs RECIST $[0.536, 0.686]$; IMmotion150: cytolytic $[0.862, 0.942]$ vs RECIST $[0.569, 0.750]$); in GSE274975 ($n = 61$) the GEP intervals touch (cytolytic $[0.777, 0.953]$ vs RECIST $[0.432, 0.787]$) while the point estimates separate cleanly. Across different cohorts the between-method spread can be as large as 0.508 (Jung: GEP 0.786 vs TIDE 0.278), so a cross-cohort comparison cannot support the claim; the controlled within-cohort comparison on Riaz is the appropriate design, because it fixes patients, treatment, and expression data and varies only the endpoint definition.

Power Analysis

Minimum detectable Δ AUROC at 80% power ($\alpha = 0.05$, two-sided) as a function of sample size and responder proportion, using the Hanley–McNeil asymptotic variance of the AUC (Mann–Whitney U) for a paired two-method comparison on the same patients (baseline AUC 0.5) with AUC correlation $r = 0.75$ — the upper end of the correlation observed between method pairs on out-of-fold scores (Supplementary Table S12). The computation is archived as `scripts/power_table_v33.py` with a machine-readable sensitivity grid over $r \in \{0, 0.5, 0.75\}$ (`power_table_v33.json`); Supplementary Table S25 reports the $r = 0.75$ (most favorable) column, rounded to 0.01, and regenerates exactly from the archived script. Weaker correlation assumptions inflate the required n roughly two- to four-fold. Full power curves in Supplementary Figure S9.

Circularity Detection Score (CDS)

CDS quantifies endpoint-feature dependency as a weighted composite: **C1 (30%)**: mean MI between endpoint-defining genes and the response label, normalized as $\min(C1_MI / 0.05, 1)$. **C2 (30%)**: mean $|\text{Spearman } \rho|$ between endpoint-defining genes and the response **C3 (40%)**: $\min(1, R/10)$, where R is the ratio of the 90th-percentile MI of the top-500-variance features to the median MI of 500 random features. The composite is $CDS = 0.3 \cdot C1 + 0.3 \cdot C2 + 0.4 \cdot C3$. These weights and normalization constants are those of the released tool (v1.2.1; source and packaging at <https://github.com/huangzhixian324-dev/cds> and in the Zenodo archive); every CDS value in this manuscript is exactly reproducible from them. One calibration caveat applies wherever a categorical label is quoted: the HIGH boundary (0.70) was calibrated on synthetic nulls, where 3% of replicates exceed it, whereas 20-36% of real-cohort random-label replicates do (Limitations); the validated reading is the continuous value against its cohort-matched null (CDS scores are identical between v1.1.0 and v1.2.1 — v1.2.1 changes only the output semantics: it adds a null-referenced reading mode, a caveats block shipping the calibration limits below, and condition-specific guidance). The CDS concept builds on earlier work by Zhao et al. (2011, BMC Genomics), who introduced a "consistency degree" metric quantifying correlation between disease endpoints and gene expression profiles [36]. We note two properties of this formulation: (a) C3 saturates at 1.0 for all evaluated endpoints and random-label replicates — highly variable genes in bulk tumor RNA-seq are predominantly immune-related, so the top-to-random MI ratio consistently hits the clipping ceiling regardless of endpoint circularity, and (b) we evaluated a re-weighting (C1 40% / C2 40% / C3 20%) as a candidate refinement, but the released tool and all reported values use the 30/30/40 formulation. Permuting C3's role (v2.0) is planned to improve MODERATE-range discrimination. **Synthetic data calibration**: To validate CDS independently of our three real endpoints, we generated synthetic expression data ($n = 200, 5,000$ genes) with controlled circularity levels (0–100% endpoint-gene signal, with 50 independent biological-signal genes as non-circular signal). CDS ranks the five scenarios monotonically ($CDS\ 0.697 < 0.820 < 0.830 < 0.832 < 0.845$; Supplementary Table S4d). Scenarios with $CDS \geq 0.80$ show inflated AUROC (0.691–0.845), while the pure-random scenario scores 0.697 — above the level of an unstructured label, reflecting C3 saturation.

Retrospective external consistency check: The Auslander et al. (2018) study provides an independent validation context for CDS. In that study, IMPRES was trained on Auslander's own data and validated externally on Hugo 2016 (RECIST endpoint), achieving AUROC = 0.83 in the original paper (0.795 in our benchmark). If CDS had been computed before that validation was performed, it would have returned

CDS = 0.449 (MODERATE under the shipped thresholds, and below its cohort-matched random-label null mean of 0.586), indicating that the Hugo RECIST endpoint is not circular — any observed AUROC reflects genuine biological signal in the IMPRES gene pairs, not endpoint-gene dependency. This counterfactual application of CDS to a published external validation scenario constitutes a retrospective consistency check: CDS's reading (non-circular endpoint → AUROC not inflated by endpoint-gene dependency) is consistent with the observed outcome (IMPRES achieved significant AUROC = 0.795 on Hugo RECIST, permutation $p = 0.0029$). The same logic applies to Lauss 2017 (ACT) (CDS = 0.781, MODERATE under the recalibrated boundary; no trainable AUROC exceeds 0.70, no inflation). For Gide 2019 no CDS value was stored at analysis time; the v33 recomputation (Supplementary Table S4b) returns CDS = 0.852 — the C1/C2 components read the strong association between the declared genes (GZMA/PRF1) and the clinical label in this large melanoma cohort as endpoint-gene dependency, i.e. the known inability of CDS v1.x to separate genuine immune-biology signal (GEP AUROC = 0.830, $q < 0.001$) from circularity. CDS correctness claims are therefore restricted to the endpoints with stored scores.

Null distribution, stress test, and reporting convention: We generated 30 independent replicates of random binary labels to estimate the CDS null distribution. Mean null CDS = 0.50 (SD = 0.09), 95th percentile = 0.65, 99th percentile = 0.71. Approximately 3% of random endpoints were flagged as HIGH under the shipped threshold (CDS > 0.70) due to C3 saturation at 1.0 — a known limitation of the current C3 formulation; no random endpoint achieved CDS < 0.30. We disclose that these 30 null replicates were generated **after** the four real benchmark endpoints had been inspected; threshold calibration is therefore retrospective, not prospective, and categorical risk labels must be treated as advisory. We deliberately ship a single threshold set (HIGH > 0.70, MODERATE 0.30–0.70, LOW ≤ 0.30) — identical in the manuscript, in the released pip package, and in all archived result files — so that users replicating our analysis obtain exactly our labels. A more conservative HIGH threshold of 0.80 (false-positive rate 0/30 on the synthetic null) is reported for transparency and will be adopted in CDS v2.0 together with a permutation-based C3. We recommend reporting the continuous CDS value; categorical risk assignments are advisory.

To characterize what CDS measures, we generated 30 pseudo-endpoints per cohort from real gene programs ^[37] — ECM/stromal (collagens, fibroblast, endothelial, pericyte and basement-membrane markers), tumor-purity/proliferation (cell cycle, E2F/MYC targets, glycolysis, oxidative phosphorylation) and lineage/signalling programs (melanocytic differentiation, keratinocyte basal, hypoxia, interferon, antigen presentation) — each dichotomized at the median of the program-mean expression, the same construction as the cytolytic surrogate, with the program's own genes declared as endpoint genes. All 60 pseudo-endpoints (Gide 2019, $n = 73$; Hugo 2016, $n = 28$) scored above the HIGH boundary (CDS 0.792–0.946): CDS flags any operationally gene-defined surrogate regardless of program identity. Random-label nulls computed per endpoint on the real feature spaces (10 replicates each; Table 3) show that the shipped 0.70 threshold also captures 3/10 null replicates at $n = 28$. CDS is therefore validated as a continuous, null-referenced measure of endpoint–gene association; the categorical HIGH label carries no specificity among structured endpoints, and the protection it enables is procedural — declaring the endpoint-defining genes identifies the circular construction itself. Full design and per-endpoint results in Supplementary Table S18 and Note S6. CDS values are reported alongside the mean, maximum and 95th percentile of a 200-replicate random-label null computed on the same cohort and feature space (Table 3); the archived 10-replicate nulls give the same readings for every endpoint. A CDS far above the null maximum indicates that the declared genes track the label beyond chance — expected for operationally gene-defined

endpoints (circularity), but also possible for genuine biology when the declared genes are true correlates of response (the Gide 2019 case, Table S4b). The two are distinguishable only by endpoint provenance, which is why the declaration of endpoint-defining genes is the primary control. **Epistemological note:** We recognize the inherent tension: CDS is itself subject to the same validation standards we recommend for benchmarking tools — prospective evaluation on independent multi-endpoint cohorts, transparent reporting of limitations, and third-party replication. Endpoint–feature overlap of the kind CDS detects is one of the leakage classes formalized in the machine-learning reproducibility literature [38]. To facilitate this, CDS is released as an open-source, independently installable package with a documented API and bundled synthetic calibration data. Its current status is experimental, and its provisional thresholds should be refined as more multi-endpoint cohorts become available.

IMPRES Implementation

IMPRES uses the canonical implementation decoded from the original authors' released feature set: 15 fixed logical relations (list in Supplementary Note S5), score = fraction fulfilled, no retraining. This matters because IMPRES on Hugo 2016 is the headline positive result of this study — it is the only FDR-significant non-circular-endpoint result at $n \leq 43$ (Supplementary Table S11); that result must rest on the published algorithm, not on an ad hoc variant. On the canonical implementation, IMPRES on Hugo 2016 yields AUROC = 0.795 (permutation $p = 0.0029$ at 50,000 prediction shuffles under the fixed-scorer protocol; $q = 0.017$ after BH). We note that Carter et al. (2019) identified potential training-set bias in the original IMPRES feature selection (34% non-random repeats) [25]. The extent to which IMPRES's Hugo performance reflects genuine biology versus inherited feature selection bias is unresolved; we report the AUROC as observed and flag this interpretative caveat.

QUANTIFICATION AND STATISTICAL ANALYSIS

Permutation testing uses full-pipeline label shuffles with frozen hyperparameters; exact p -values are $(ge + 1)/(n + 1)$. Within-cohort multiple testing uses Benjamini-Hochberg FDR ($q < 0.05$). AUROC 95% CIs use the paired bootstrap (1,000 iterations, the convention used throughout and in Table S9); correlated-AUROC comparisons likewise use the paired bootstrap rather than DeLong. Power analysis and minimum-sample guidance are derived from observed effect sizes (Supplementary Table S9).

ADDITIONAL RESOURCES

None.

KEY RESOURCES TABLE

Resource	Source	Identifier
SPATBench pipeline (v33)	This paper; GitHub (https://github.com/huangzhixian324-dev/spa-bench)	spa-bench:v33
circularity-detection-score	This paper; GitHub (https://github.com/huangzhixian324-dev/cds)	v1.2.1
SPATBench archive (Zenodo)	This paper	10.5281/zenodo.22746254
tidepy	Jingxin Fu	v1.3.9
scikit-learn	scikit-learn.org	>=1.0
GSE78220 / GSE91061 / GSE100797 / GSE135222	GEO	GSE78220; GSE91061; GSE100797; GSE135222
mel/blca/liu iatlas cohorts	cBioPortal	mel_iatlas_gide_2019; blca_iatlas_invigor210_2017; mel_iatlas_liu_2019

SUPPLEMENTARY DATA

Supplementary Data are provided as a separate PDF document (SPATBench_CellSystems_supplementary_v35.pdf; Markdown source supplementary_material_v33.md, generated by scripts/synthesize_v33_supplementary.py from the machine-readable outputs in results/benchmark/v33/). Contents:

- **Table S1:** Per-fold cross-validation setup
- **Table S2:** Literature comparison — published vs. SPATBench AUROC
- **Table S3:** Cohort characteristics (detailed)
- **Table S4:** CDS — full component breakdown, synthetic calibration data, null distribution analysis, recalibrated thresholds
- **Table S5:** Method concordance — Cohen's κ , full 10-pair matrix (Hugo 2016)
- **Table S6:** IMPRES leave-one-pair-out analysis
- **Table S7:** Preprocessing sensitivity
- **Table S8:** Cross-validation sensitivity
- **Table S9:** Full five-cohort benchmark results (AUROC/CI/AUPRC, clinical-utility metrics, survival stratification)
- **Table S10:** Immune deconvolution marker genes (43 genes, 14 signatures)
- **Table S11:** Complete permutation test results (all 36 cells, including the frozen-pipeline observed AUROC)
- **Table S12:** Random seed stability analysis (10 seeds)
- **Table S13:** Learning curve analysis (Hugo 2016, $n = 8-28$)
- **Table S14:** Effect-size comparison — Riaz 2017
- **Table S15:** TCGA Cox regression (12 signatures $\times$ 4 cancer types, 2,491 patients; COAD re-acquired from the GDC current index)
- **Table S16:** k-fold robustness analysis — Riaz RECIST ($k = 2, 3, 5$)
- **Table S17:** CDS retrospective assessment on five published benchmark studies + computed CDS for the iAtlas-accessible cohorts (E3)
- **Table S18:** CDS negative-control stress test — 60 real gene-program pseudo-endpoints (E2)

- **Table S19:** External replication — response-definition effect on Liu 2019 (independent cohort, E1)
- **Table S20:** Cross-cancer replication — IMvigor210 (bladder carcinoma, non-melanoma, E1)
- **Table S21:** Multiple-comparison family sensitivity — BH q under within-cohort / global / method-class-stratified family definitions (36 cells)
- **Table S22:** AUROC across four clinical-endpoint cohorts (migrated from the main text; 95% CIs where shown)
- **Table S23:** DCB endpoint, Jung 2019 (n = 27, NSCLC; corrected labels)
- **Table S24:** Overall survival stratification on Hugo 2016 (true events)
- **Table S25:** Minimum detectable Δ AUROC at 80% power (Hanley–McNeil paired model, $r = 0.75$)
- **Table S26:** Clinical–clinical response-definition contrast on Liu 2019 (RECIST vs PFS-derived DCB)
- **Table S27:** Executed NSCLC replication — GSE274975 (Poddubskaya 2024)
- **Table S28:** Fourth-cancer-type replication — IMmotion150 ccRCC (McDermott 2018)
- **Figures S1–S12**
- **Notes S1–S6**

FIGURE LEGENDS

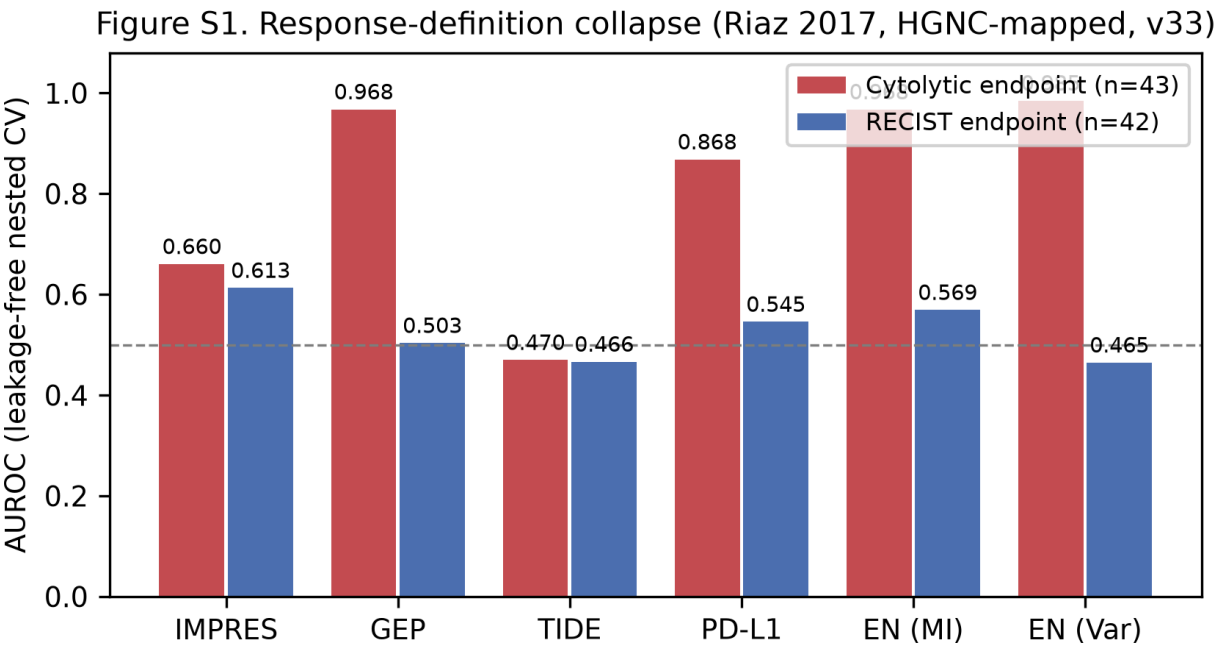


Figure 1. Response-definition collapse on Riaz 2017. AUROC per method on the same patients under the cytolytic-surrogate endpoint versus the clinical RECIST endpoint (HGNC-mapped matrix, v33; 95% bootstrap CIs). Every method whose feature set overlaps the endpoint-defining genes (GZMA/PRF1) collapses; IMPRES, with no overlap, is nearly insensitive. The single FDR-significant clinical-endpoint cell (IMPRES on Hugo, 0.795) carries the Carter et al. training-set-bias caveat stated in Methods and Limitations and should not be read as validated biology until that question is resolved. The identical file is provided as Supplementary Figure S1.

Alt text: Bar chart comparing AUROC of six prediction methods on the same Riaz 2017 patients under two endpoints: the gene-defined cytolytic surrogate (all methods with overlapping feature sets score high,

0.83-0.99) versus the clinical RECIST endpoint (all collapse to 0.40-0.57). IMPRES, whose feature set does not overlap the endpoint genes, is nearly unchanged between endpoints.

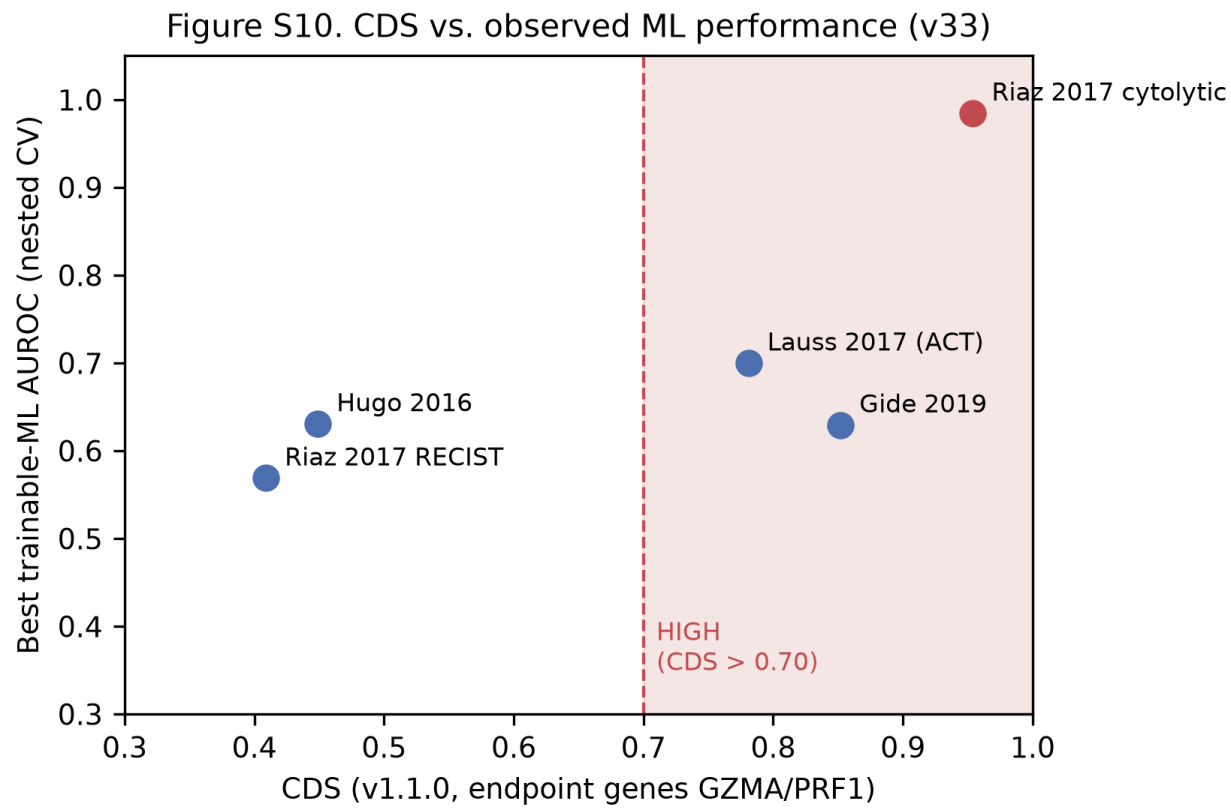


Figure 2. CDS versus observed trainable-ML performance. Circularity Detection Score (v1.2.1; scores identical to the v1.1.0 recomputation, Table S4b) against the best trainable-ML AUROC per endpoint, with cohort-matched random-label nulls; the circular cytolitic endpoint separates from clinical endpoints. The identical file is provided as Supplementary Figure S10.

Alt text: Scatter plot of Circularity Detection Score versus best trainable-ML AUROC across five endpoints, with cohort-matched random-label nulls. The circular cytolitic endpoint (CDS 0.954) separates from the clinical endpoints, which sit at or inside their nulls except Gide RECIST (CDS 0.852, above its null maximum, reflecting genuine biology the categorical label cannot separate).

Figure S12. Complete permutation matrix (E4): observed AUROC + p per cell; * BH $q < 0.05$ within cohort. Fixed scorers: 50,000 prediction shuffles; ML: full-pipeline shuffles (per-cell n); n.p. = not permuted (PD-L1 excluded in v33 runs)

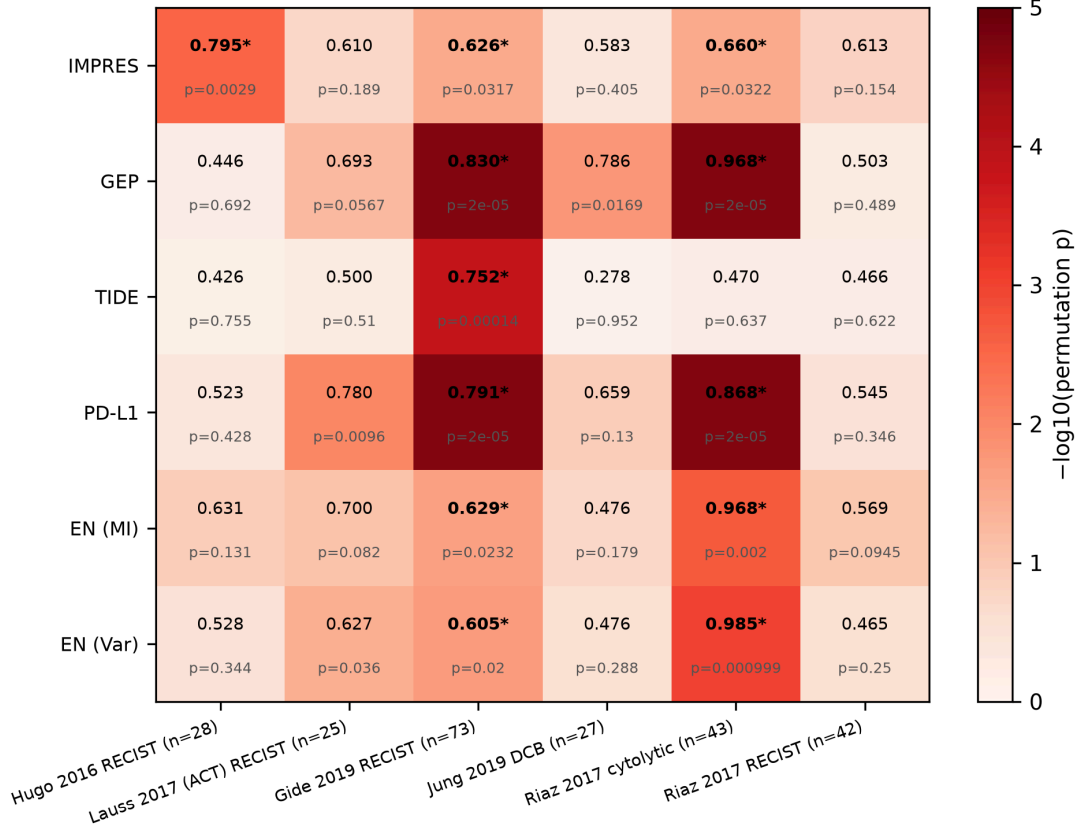


Figure 3. Complete permutation test matrix. Observed AUROC and permutation p per method–cohort cell; asterisks mark within-cohort BH $q < 0.05$. No trainable method survives FDR on any clinical endpoint at $n \leq 43$. The identical file is provided as Supplementary Figure S12.

Alt text: Heatmap of the complete 36-cell permutation test matrix showing observed AUROC and permutation p per method-cohort cell; asterisks mark within-cohort BH q below 0.05. No trainable method survives FDR correction on any clinical endpoint with n at most 43.

ACKNOWLEDGEMENTS

The author thanks the maintainers of the public datasets used in this study.

AUTHOR CONTRIBUTIONS

The sole author was responsible for all aspects of this work: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review and editing, Visualization, Supervision, Project administration, and Funding acquisition.

FUNDING

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

DECLARATION OF INTERESTS

The author declares no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the author used ChatGPT (OpenAI) in order to improve the clarity and language of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

REFERENCES

1. Topalian, S.L., Hodi, F.S., Brahmer, J.R., Gettinger, S.N., Smith, D.C., McDermott, D.F., Powderly, J.D., Carvajal, R.D., Sosman, J.A., Atkins, M.B., et al. (2012). Safety, activity, and immune correlates of anti-PD-1 antibody in cancer. *N Engl J Med*, 366, 2443–2454. doi:10.1056/NEJMoa1200690
2. Hodi, F.S., O'Day, S.J., McDermott, D.F., Weber, R.W., Sosman, J.A., Haanen, J.B., Gonzalez, R., Robert, C., Schadendorf, D., Hassel, J.C., et al. (2010). Improved survival with ipilimumab in patients with metastatic melanoma. *N Engl J Med*, 363, 711–723. doi:10.1056/NEJMoa1003466
3. Ayers, M., Lunceford, J., Nebozhyn, M., Murphy, E., Loboda, A., Kaufman, D.R., Albright, A., Cheng, J.D., Kang, S.P., Shankaran, V., et al. (2017). IFN- γ -related mRNA profile predicts clinical response to PD-1 blockade. *J. Clin. Invest.* 127, 2930–2940. doi:10.1172/JCI91190
4. Jiang, P., Gu, S., Pan, D., Fu, J., Sahu, A., Hu, X., Li, Z., Traugh, N., Bu, X., Li, B., et al. (2018). Signatures of T cell dysfunction and exclusion predict cancer immunotherapy response. *Nat. Med.* 24, 1550–1558. doi:10.1038/s41591-018-0136-1
5. Auslander, N., Zhang, G., Lee, J.S., Frederick, D.T., Miao, B., Moll, T., Tian, T., Wei, Z., Madan, S., Sullivan, R.J., et al. (2018). Robust prediction of response to immune checkpoint blockade therapy in metastatic melanoma. *Nat. Med.* 24, 1545–1549. doi:10.1038/s41591-018-0157-9
6. Chowell, D., Yoo, S.K., Valero, C., Pastore, A., Krishna, C., Lee, M., Hoen, D., Shi, H., Kelly, D.W., Patel, N., et al. (2022). Improved prediction of immune checkpoint blockade efficacy across multiple cancer types. *Nat Biotechnol*, 40, 499–506. doi:10.1038/s41587-021-01070-8
7. Riaz, N., Havel, J.J., Makarov, V., Desrichard, A., Urba, W.J., Sims, J.S., Hodi, F.S., Martín-Algarra, S., Mandal, R., Sharfman, W.H., et al. (2017). Tumor and microenvironment evolution during immunotherapy with nivolumab. *Cell* 171, 934–949.e16. doi:10.1016/j.cell.2017.09.028
8. Hugo, W., Zaretsky, J.M., Sun, L., Song, C., Moreno, B.H., Hu-Lieskovan, S., Berent-Maoz, B., Pang, J., Chmielowski, B., Cherry, G., et al. (2016). Genomic and transcriptomic features of response to anti-PD-1 therapy in metastatic melanoma. *Cell* 165, 35–44. doi:10.1016/j.cell.2016.02.065

9. Simon, R., Radmacher, M.D., Dobbin, K. and McShane, L.M. (2003). Pitfalls in the use of DNA microarray data for diagnostic and prognostic classification. *J Natl Cancer Inst*, 95, 14–18. doi:10.1093/jnci/95.1.14
10. Vabalas, A., Gowen, E., Poliakoff, E. and Casson, A.J. (2019). Machine learning algorithm validation with a limited sample size. *PLoS One* 14, e0224365. doi:10.1371/journal.pone.0224365
11. Button, K.S., Ioannidis, J.P.A., Mokrysz, C., Nosek, B.A., Flint, J., Robinson, E.S.J. and Munafò, M.R. (2013). Power failure: why small sample size undermines the reliability of neuroscience. *Nat. Rev. Neurosci.* 14, 365–376. doi:10.1038/nrn3475
12. Shi, L., Campbell, G., Jones, W.D., Campagne, F., Wen, Z., Walker, S.J., Su, Z., Chu, T.M., Goodsaid, F.M., Pustai, L., et al. (2010). The MicroArray Quality Control (MAQC)-II study of common practices for the development and validation of microarray-based predictive models. *Nat. Biotechnol.* 28, 827–838. doi:10.1038/nbt.1665
13. Berrar, D. and Flach, P. (2012). Caveats and pitfalls of ROC analysis in clinical microarray research (and how to avoid them). *Brief. Bioinform.* 13, 83–97. doi:10.1093/bib/bbr008
14. Rooney, M., Shukla, S., Wu, C., Getz, G. and Hacohen, N. (2015). Molecular and genetic properties of tumors associated with local immune cytolytic activity. *Cell* 160, 48–61. doi:10.1016/j.cell.2014.12.033
15. Liang, Y., Chhuo, L., Argha, A., Farbehi, N., Chen, L., Alizadehsani, R., Hosseinzadeh, M., Beheshti, A., Porntaveetus, T., Ye, Y., et al. (2026). Transcriptomic models for immunotherapy response prediction show limited cross-cohort generalisability. *arXiv* 2604.05478 [q-bio.GN]. doi:10.48550/arXiv.2604.05478
16. Shen, W., Moon, I., Nguyen, T.H., Li, M.M., Huang, Y., Nair, N., Marbach, D. and Zitnik, M. (2026). Generalizable AI predicts immunotherapy outcomes across cancers and treatments. *Nat. Med.* 32, 3010–3022. doi:10.1038/s41591-026-04502-7
17. El Kanbi, K., Cattani, Y., Marschall, P. et al. (2026). A transcriptomic benchmark for foundation models in immunology and inflammation drug development. *Learning Meaningful Representations of Life (LMRL) Workshop at ICLR 2026* (spotlight).
18. Pal, L.R., Gertz, E.M., Nair, N.U., Mukherjee, S., Patiyal, S., Cantore, T., Campagnolo, E.M., Chang, T.G., Dhruva, S.R., Kim, Y., et al. (2025). A machine learning framework for supervised treatment response prediction from tumor transcriptomics: a large-scale pan-cancer study. *bioRxiv*. doi:10.1101/2025.10.24.684491
19. Zhou, Q., Kirshtein, A. and Shahriyari, L. (2026). Towards the tumor microenvironment scoring methods for immune checkpoint inhibitor response. *npj Precis. Oncol.*, 10, 88. doi:10.1038/s41698-025-01221-z
20. Liu, D., Schilling, B., Liu, D., Sucker, A., Livingstone, E., Jerby-Arnon, L., Zimmer, L., Gutzmer, R., Satzger, I., Loquai, C., et al. (2019). Integrative molecular and clinical modeling of clinical outcomes to PD1 blockade in patients with metastatic melanoma. *Nat. Med.* 25, 1916–1927. doi:10.1038/s41591-019-0654-5
21. Poddubskaya, E., Suntsova, M., Lyadova, M., Luppov, D., Guryanova, A., Lyadov, V., Garazha, A., Sorokin, M., et al. (2024). Biomarkers of success of anti-PD-(L)1 immunotherapy for non-small cell lung cancer derived from RNA- and whole-exome sequencing: results of a prospective observational study on a cohort of 85 patients. *Front. Immunol.* 15, 1493877. doi:10.3389/fimmu.2024.1493877
22. Mariathasan, S., Turley, S.J., Nickles, D., Castiglioni, A., Yuen, K., Wang, Y., Kadel, E.E., III, Koeppen, H., Astarita, J.L., Cubas, R., et al. (2018). TGFβ attenuates tumour response to PD-L1 blockade by contributing to exclusion of T cells. *Nature* 554, 544–548. doi:10.1038/nature25501

23. McDermott, D.F., Huseni, M.A., Atkins, M.B., Motzer, R.J., Rini, B.I., Escudier, B., Fong, L., Joseph, R.W., Pal, S.K., et al. (2018). Clinical activity and molecular correlates of response to atezolizumab alone or in combination with bevacizumab versus sunitinib in advanced renal cell carcinoma. *Nat. Med.* 24, 749–757. doi:10.1038/s41591-018-0053-3
24. Kim, S.T., et al. (2022). Metastatic gastric cancer treated with anti-PD-1 immunotherapy (PRJEB25780). RNA-seq data from the European Nucleotide Archive and the TIGER portal.
25. Carter, J.A., Gilbo, P. and Atwal, G.S. (2019). IMPRES does not reproducibly predict response to immune checkpoint blockade therapy in metastatic melanoma. *Nat Med*, 25, 1833–1835. doi:10.1038/s41591-019-0671-4
26. Usset, J., Rosendahl Huber, A., Andrianova, M.A., Batlle, E., Carles, J., Cuppen, E., Elez, E., Felip, E., Gómez-Rey, M., Lo Giacco, D., et al. (2024). Five latent factors underlie response to immunotherapy. *Nat. Genet.* 56, 2112–2120. doi:10.1038/s41588-024-01899-0
27. Efron, B. and Tibshirani, R.J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall, New York.
28. Van Allen, E.M., Miao, D., Schilling, B., Shukla, S.A., Blank, C., Zimmer, L., Sucker, A., Hillen, U., Geukes Foppen, M.H., Goldinger, S.M., et al. (2015). Genomic correlates of response to CTLA-4 blockade in metastatic melanoma. *Science*, 350, 207–211. doi:10.1126/science.aad0095
29. Lauss, M., Donia, M., Harbst, K., Andersen, R., Mitra, S., Rosengren, F., Salim, M., Vallon-Christersson, J., Törngren, T., Kvist, A., et al. (2017). Mutational and putative neoantigen load predict clinical benefit of adoptive T cell therapy in melanoma. *Nat. Commun.* 8, 1738. doi:10.1038/s41467-017-01460-0
30. Jung, H., Kim, H.S., Kim, J.Y., Sun, J.M., Ahn, J.S., Ahn, M.J., Park, K., Esteller, M., Lee, S.H. and Choi, J.K. (2019). DNA methylation loss promotes immune evasion of tumours with high mutation and copy number load. *Nat. Commun.* 10, 4278. doi:10.1038/s41467-019-12159-9
31. Gide, T.N., Quek, C., Menzies, A.M., Tasker, A.T., Shang, P., Holst, J., Madore, J., Lim, S.Y., Velickovic, R., Wongchenko, M., et al. (2019). Distinct immune cell populations define response to anti-PD-1 monotherapy and anti-PD-1/anti-CTLA-4 combined therapy. *Cancer Cell* 35, 238–255.e6. doi:10.1016/j.ccell.2019.01.003
32. Goodman, A.M., Kato, S., Bazhenova, L., Patel, S.P., Frampton, G.M., Miller, V., Stephens, P.J., Daniels, G.A. and Kurzrock, R. (2017). Tumor mutational burden as an independent predictor of response to immunotherapy in Diverse Cancers. *Mol Cancer Ther*, 16, 2598–2608. doi:10.1158/1535-7163.MCT-17-0386
33. Newman, A.M., Liu, C.L., Green, M.R., Gentles, A.J., Feng, W., Xu, Y., Hoang, C.D., Diehn, M. and Alizadeh, A.A. (2015). Robust enumeration of cell subsets from tissue expression profiles. *Nat. Methods* 12, 453–457. doi:10.1038/nmeth.3337
34. Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc B*, 57, 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x
35. DeLong, E.R., DeLong, D.M. and Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44, 837–845. doi:10.2307/2531595
36. Zhao, C., Shi, L., Tong, W., Shaughnessy, J.D., Oberthuer, A., Pusztai, L., Deng, Y., Symmans, W.F. and Shi, T. (2011). Maximum predictive power of the microarray-based models for clinical outcomes is limited by correlation between endpoint and gene expression profile. *BMC Genomics* 12(Suppl 5), S3. doi:10.1186/1471-2164-12-S5-S3

- 1102 37. Subramanian, A., Tamayo, P., Mootha, V.K., Mukherjee, S., Ebert, B.L., Gillette, M.A., Paulovich,
1103 A., Pomeroy, S.L., Golub, T.R., Lander, E.S., et al. (2005). Gene set enrichment analysis: a
1104 knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci.*
1105 *U.S.A.* 102, 15545–15550. doi:10.1073/pnas.0506580102
- 1106 38. Kapoor, S. and Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-
1107 based science. *Patterns* 4, 100804. doi:10.1016/j.patter.2023.100804